Chapter 1

INTRODUCTION TO RATIONAL USE OF MEDICINES

Rational Use of Medicines requires that patients receive medications appropriate to their clinical needs, in doses that meet their own individual requirements, for an adequate period of time, and at affordable cost to them and the community (WHO, 1985). In other words, rational use means 'prescribing right medicines, in adequate dose for the sufficient duration and appropriate to the clinical needs of the patient at lowest cost'. Medicine use is a complex cyclical process and in the clinical context, rational drug use implies correct drug, appropriate indication and selection with respect to efficacy, safety, suitability for the patient and cost. The concept of *Rational Use of Medicines* is age old; however, it has attained more significance and momentum in terms of medical and socioeconomic aspects recently. Three tools have been identified to implement *Rational Use of Medicines* effectively. These include Essential Medicines List, Standard Treatment Guidelines and Drug Formulary. To be meaningful, all three tools should work in concert and subject for periodic review and update.

I. The concept of Essential Medicines

Essential Medicines (EM) are those that satisfy the health care needs of the majority of the population and therefore should be available at all times in appropriate amounts and in appropriate dosage forms. Following the concept of EM, WHO published the first Essential Medicine List (EML) in 1977, which has been updated successively every 5 year in view of changing world scenario about the medical needs and availability of new and better drugs. The selection of these medicines is based on public health relevance, evidence of efficacy and safety and comparative cost-effectiveness. These priority medicines should be available at each functioning health care facility with assured quality and adequate information at a price the individual and the community can afford.

The selection of essential medicines depends upon level of health care facility, adequate data on the efficacy, safety, suitability and comparative cost effectiveness of available treatments. In addition, stability of the product, need for special diagnostic or treatment facilities and pharmacokinetic properties are also considered. Most essential medicines are formulated as single compound. However, fixed dose combinations are selected only when it has proven pharmacokinetic and/or pharmacodynamic advantages over single compound administered separately. While comparing the cost of medicines among the alternative treatments, the cost of total treatment and not the unit cost is considered. Other factors taken into account include pattern of diseases, treatment facilities, training and experience of the available personnel, financial resources etc.

Advantages of Essential Medicines

The implementation of the concept of EM has several logistic and prescribing advantages. For example better management of medicines i.e. easier procurement, storage,

distribution, more manageable stock, better quality assurance, easier dispensing, less fragmentation of budget with improve drug availability. Moreover, prescribing improves due to focused training and drug information, better recognition of adverse drug reactions, drug interactions and focused education efforts. The list of EM closely relates to treatment guidelines for clinical diseases and is also used for the procurement and supply of medicines in public sector, schemes for reimburses medicines costs and local medicine production.

Its implementation is intended to be flexible and adaptable to different situations; which medicines are regarded as essential remains a national responsibility.

II. Standard Treatment Guidelines

Standard Treatment Guidelines (STGs), treatment protocols, and prescribing policies are defined as systematically developed statements designed to assist prescribers in making decisions about the appropriate treatment and health care for specific clinical problem. These guidelines usually reflect the consensus on the optimal treatment options within a health facility or system. In fact, they represent one of the approaches to promote therapeutically effective and economically efficient use of medicines.

STGs list the preferred pharmacological and non-pharmacological treatment for common health problems experienced by people in a specific health system. The pharmacological treatment should include the name, dosage form, strength, average dose (pediatric and adult), number of doses per day, and duration of treatment. Other information on diagnosis and advice to the patient may also be included. The most common diseases and serious medical conditions that have high morbidity and mortality rate are selected. Treatment guidelines must have the most up-to-date and evidence based information to determine appropriate treatment options by expert authors and reviewers. Effective implementation, however, is perhaps the greatest challenge in introducing STGs. When implemented effectively, it offers several advantages to patients, health care providers, supply management personnel and policy makers as follows:

Patients,

- Most cost-effective optimal therapy is provided.
- Better compliance through consistency among prescribers.
- Improvement in availability of drugs.
- Overall better quality of care.

Health care providers,

- Provides standardized guidance to practitioners
- Quality of care standard- based on expert consensus and basis for monitoring
- Utilizes only formulary or essential medicines, so the health care system needs to provide only the medicines in the STGs.

Supply managers,

- Identification of medicines to be available for the most commonly treated problems.

- Facilitation of prepackaging of course-of-therapy quantities of commonly prescribed items.
- Medicine demand more predictable, so forecasting more reliable

Health care policy makers,

- Provides focus therapeutic integration of special programs (diarrhea disease control, acute respiratory infection)
- Provides a system for controlling cost by using funds more efficiently

STGs also have certain drawbacks. Establishing, developing and implementation of STGs is a difficult, lengthy and time consuming process. Moreover, it needs to be updated regularly to reflect changes in treatment strategies. If a regular schedule for updating the STGs is not followed, they lose credibility and become obsolete. The extent of information to be provided in each medical condition is often a difficult decision. However, the STG manual should be concise and small enough to carry it easily.

Key features of a successful STG manual

Simplicity- The number of health problem is limited to commonly observed conditions. Each clinical condition list a few salient features with clear and concise information on pharmacological and non- pharmacological treatment.

Credibility- Guidelines developed by the most respected experienced clinicians in the country and revision is based on actual experience.

Treatment standards for all levels of health care facility- Health care providers use the same standard treatment; however, the referral criterion differs. While the first choice treatment for a patient depends upon the patient's diagnosis and condition and not on the prescriber.

Medicine supply based on standards- The supply of medicines must match with the standard treatment and list of essential medicines.

Regular updating- Any change in the therapeutic options, bacterial resistance pattern should be incorporated in the revised version to reflect current recommendations.

User friendly- The STGs should be published as small, durable pocket manual which make it convenient to carry and use.

III. Drug Formulary

A formulary is a compendium of practical drug information for clinical practice adapted for use at a particular health care facility. It is a ready reference to key information on the selection, prescribing, dispensing and administration of medicines. Although the concept of formulary is old, a drug formulary is considered an essential element in achieving rational prescribing in addition to controlling the costs. The task of development of a formulary in a hospital is usually undertaken by Drugs and Therapeutics Committee with a multidisciplinary membership of doctors, pharmacists, nurses and administrative staff.

The consultants place a request for inclusion of medicines along with its dosage formulations to Drugs and Therapeutics Committee. The committee may accept the request and include it in formulary or reject and suggest an alternative with information to the prescriber. The committee then monitors the impact, cost and benefits of various medicines in the health care facility. Depending on the outcome, the use of medicine is extended or curtailed. Formularies have significant advantages for patients care. However, the introduction of formularies may cause friction since it may be viewed as a major threat to their clinical freedom. Effective multidisciplinary working methods and consultation is needed to overcome this attitude. Potential advantages and disadvantages of drug formularies are as follows:

Advantages

- Reduces range of products to be purchased / stored / distributed
- Encourages generic prescribing
- Lowers drug costs
- Improved knowledge of a limited number of medicines
- Improves drug safety
- Improves patient care
- Provides opportunity for multidisciplinary work

Disadvantages

- Restricts prescribing choice
- Reduces treatment options
- Inhibits introduction of new products
- Restricts research efforts

IV. Rational Approach To Therapeutics

Medicines should only be prescribed when necessary, and the benefit of administering the medicine should be considered in relation to the risks involved. Bad prescribing habits lead to ineffective and unsafe treatment, exacerbation or prolongation of illness, distress and harm to the patient, and higher cost. *The Guide to Good Prescribing;* WHO; 1994, provides medical undergraduates with important tools for training in the process of rational prescribing.

The following steps will help to remind prescribers of the rational approach to therapeutics.

1. Define the patient's problem

Whenever possible, making the right diagnosis is based on integrating many pieces of information: the complaint as described by the patient; a detailed history; physical examination; laboratory tests; X-rays and other investigations. This will help in rational prescribing, always bearing in mind that disease are evolutionary process.

2. Specify the therapeutic objective

Doctors must clearly state their therapeutic objectives based on the pathophysiology underlying the clinical situation. Very often physicians must select more than one therapeutic goal for each patient.

3. Selecting therapeutic strategies

The selected strategy should be discussed with the patient. The selected treatment can be non-pharmacological and/or pharmacological; it also needs to take into account the total cost of all therapeutic options.

a. Non-pharmacological treatment

The patient does not always need a medicine for treatment of the condition. Very often, health problems can be resolved by a change in life style, dietary habits, use of physiotherapy or exercise, provision of adequate psychological support and other non-pharmacological treatments; they have the same importance as a prescription drug and instructions must be written, explained and monitored in the same way.

b. Pharmacological treatment

Selecting the correct group of drugs

Knowledge about the pathophysiology involved in the clinical situation of each patient and the pharmacodynamics of the chosen group of drugs, are two of the fundamental principles for rational therapeutics.

Selecting the drug from the chosen group

The selection process must consider benefit/risk/cost information. This step is based on evidence about maximal clinical benefits of the drug for a given indication (efficacy) with minimum production of adverse effects (safety).

It must be remembered that each drug has adverse effects and it is estimated that up to 10% of hospital admissions in industrialized countries are due to adverse effects. Not all drug-induced injury can be prevented but much of it is caused by inappropriate selection of drugs.

In cost comparisons between drugs, the cost of total treatment and not only the unit cost of the drug must be considered.

Verifying the suitability of the chosen pharmaceutical treatment of each patient

The prescriber must check suitability of the chosen active substance, its dosage form, dosage schedule and duration of treatment are suitable for each patient. Drug treatment should be individualized to the needs of each patient.

Prescription writing

The prescription is the link between the prescriber, the pharmacist (or dispenser) and the patient so it is important for successful management of presenting medical condition. The item is covered in more detail in the following section.

Giving information, instructions and warnings

This step is important to ensure patient adherence and is covered in detail in the following section.

Monitoring treatment

Evaluation of the follow up and the outcome of treatment provides information whether the patient's problem is solved or to reformulate, if necessary. This step gives rise to important information about the effects and adverse effects of medicines, contributing to the evidence of knowledge, necessary to promote the rational use of drugs.

V. Adverse Drug Effects and Drug Interactions

Adverse drug reactions

An adverse drug reaction (ADR) may be defined as '*A response to a drug which is noxious, unintended and occurs at doses normally used for prophylaxis, diagnosis or therapy of disease, or for the modification of physiological function'*. ADRs are therefore unwanted or unintended effects of a medicine, including idiosyncratic effects, which occur during its proper use.

An adverse drug event (ADE) is defined as '*Any untoward medical occurrence that may present during treatment with a medicine but which does not necessarily have a causal relationship with this treatment'.*

Two types of ADRs and their differentiating characteristics have been identified. The common, predictable, dose related reaction with low mortality is type 'A' ADR. For example, anticholinergic effect of tricyclic antidepressants and digoxin toxicity. ADRs may also be unrelated to the known pharmacological action of a drug. These are uncommon, non-dose related bizarre 'B' type reactions, for example, anaphylaxis with penicillins, malignant hyperthermia with halothane.

Thalidomide marked the first recognized public health disaster related to the introduction of a new drug. It is now recognized that clinical trials, however through, cannot be guaranteed to detect all adverse effects likely to be caused by a drug. Health workers are thus encouraged to record and report to their national pharmacovigilance centre any unexpected adverse effects with any drug to achieve faster recognition of serious related problems. The Central Drugs Standard Control Organization (CDSCO), Director General of Health Services under the influence of Ministry of Health & Family Welfare, has initiated a nationwide Pharmacovigilance programme for protecting the health of the patients by assuring drug safety.

The specific aims and objectives of the National Pharmacovigilance Programme are:

- To monitor ADRs in Indian population.
- To create awareness amongst health care professionals about the importance of ADR reporting in India
- To ensure that the benefits of use of medicine outweighs the risks and thus safeguard the health of the Indian population
- To generate independent, evidence based recommendations on the safety of medicines

- To encourage safe, rational and cost effective use of medicines.

Under the National Coordinating Center, several ADR monitoring centers operates to detect ADEs. These ADR monitoring centers are located at MCI approved medical colleges and hospitals, private hospitals, public health programmes, and autonomous institutions such as Indian Council of Medical Research etc. The ADR monitoring centers record and analyze the ADEs and submit them to the National Coordinating center via online reporting system known as Vigiflow.

Spontaneous reporting

Spontaneous reporting is the reporting of ADRs by healthcare professionals (and in some countries consumers) to identify and report any suspected adverse drug reaction to a pharmacovigilance center and/or to the manufacturer. It is one of the important methods of collecting information about ADRs.

All suspected adverse reactions, known or unknown, serious or otherwise should be reported. It is important to report even minor reactions of new drugs. For established drugs, the reporting of serious ADRs is important. An increased frequency of a known ADR to a drug is also important.

Adverse drug reactions may be reported or notified to the nearest ADR monitoring centre under the National Pharmacovigilance Program. The Department of Pharmacology, B. J. Medical College is one of the approved ADR monitoring center in Gujarat state. The ADRs can be reported by the forms available at the department. Reporting is the process of providing ADR information by filling in the ADR form appropriately giving details of the clinical case as well as the ADR (Appendix – I).

Major factors predisposing to adverse drug effects

It is well known that different patients often respond differently to a given treatment regimen. For example, patients taking combinations of drugs known to interact, only a small number show any clinical evidence of interactions. In addition to the pharmaceutical properties of drug therefore, there are characteristics of the patient which predispose to ADRs.

Extremes of age: The very old and the very young are more susceptible to ADRs. Drugs which commonly cause problems in the elderly include hypnotics, diuretics, non-steroidal anti-inflammatory drugs, antihypertensives, psychotropics and digoxin.

All children and particularly neonates, differ from adults in their response to drugs. Some drugs are likely to cause problems in neonates (for example morphine), but are generally tolerated in children. Other drugs (for example valproic acid) are associated with increased risk of ADRs in children of all ages. Other drugs associated with problems in children include chloramphenicol (Grey baby syndrome), antiarrhythmics (worsening of arrhythmias), acetylsalicylic acid (Reye syndrome).

Intercurrent illness: If besides the condition being treated, the patient suffers from another disease, such as kidney, liver or heart disease, special precautions may be

necessary to prevent ADRs. The genetic make-up of the individual patient may also predispose to ADRs.

Drug interactions: May occur between drugs which compete for the same receptor or act on the same physiological system. They may also occur indirectly when a drug-induced disease or a change in fluid or electrolyte balance alters the response to another drug.

Incompatibilities between drugs and intravenous fluids

Drugs should not be added to blood, amino acid solutions or fat emulsions. Certain drugs, when added to intravenous fluids, may be inactivated by pH changes, by precipitation or chemical reaction. Benzylpenicillin and ampicillin lose potency after 6-8 hours if added to dextrose solutions, due to the acidity of the solutions. Some drugs bind to plastic containers and tubing, for example diazepam and insulin. Aminoglycosides are incompatible with penicillins and heparin. Hydrocortisone is incompatible with heparin, tetracycline and chloramphenicol.

Adverse effects caused by traditional medicines

Patients who have been or are taking traditional herbal remedies may develop ADRs. It is not always easy to identify the responsible plant or plant constituent. Refer to the drug and toxicology information service if available or to suitable literature.

The effect of food on drug absorption

Food delays gastric emptying and reduces the rate of absorption of many drugs; the total amount of drug absorbed may or may not be reduced. However, some drugs are preferably taken with food, either to increase absorption or to decrease the irritant effect on the stomach.

VI. Prescription writing

A prescription is an instruction from a prescriber to a dispenser. Every country has its own standards for the minimum information required for a prescription, and its own laws and regulations to define which drugs require a prescription and who is entitled to write it. Many countries have separate regulations for prescriptions for controlled drugs such as opioid analgesics. The following guidelines will help to ensure that prescriptions are correctly interpreted and leave no doubt the intension of the prescriber. The guidelines are relevant for primary care prescribing; they may, however, be adapted for use in hospitals or other specialist units.

Prescription form

The most important requirement is that the prescription be ***clear and legible*** and indicate precisely what should be given. The local language may be preferred.

The following details should be shown on the form:

- The prescriber's name, address and telephone number. This will allow either the patient or the dispenser to contact the prescriber for any clarification or potential problem with the prescription.

- Date of the prescription. In many countries the validity of a prescription has no time limit, but in some countries pharmacists do not dispense drugs on prescriptions older than 3 to 6 months.
- Name, form and strength of the drug. The International Non-proprietary Name of the drug should always be used. If there is a specific reason to prescribe a special brand, the trade name can be added. Generic substitution is allowed in some countries. The pharmaceutical form (for example 'tablet', 'oral solution', 'eye ointment') should also be stated.
- The strength of the drug should be stated in standard units using abbreviations that are consistent with the System International (SI). 'Microgram' and 'nanogram' should not, however, be abbreviated. Also, 'units' should not be abbreviated. Avoid decimals whenever possible. If unavoidable, a zero should be written in front of the decimal point.
- Specific areas for filling in details about the patient including name, address and age.

Directions

Directions specifying the route, dose and frequency should be clear and explicit; use of phrases such as 'take as directed' or 'take as before' should be avoided.

For preparations which are to be taken on an 'as required' basis, the minimum dose interval should be stated together with, where relevant, the maximum daily dose. It is good practice to qualify such prescriptions with the purpose of the medication (for example 'every 6 hours as required for pain', 'at night as required to sleep').

It is good practice to explain the directions to the patient; these directions will then be reinforced by the label on the medicinal product and possibly by appropriate counseling by the dispenser. It may be worthwhile giving a written note for complicated regimens although it must be borne in mind that the patient may lose the separate note.

Quantity to be dispensed

The quantity of the medicinal product to be supplied should be stated such that it is not confused with either the strength of the product or the dosage directions.

Alternatively, the length of the treatment course may be stated (for example 'for 5 days').

Wherever possible, the quantity should be adjusted to match the pack sizes available.

For liquid preparations the quantity should be stated in milliliters (abbreviated as 'ml') or litres (preferably not abbreviated since the letter 'l' could be confused with the figure '1').

Narcotics and controlled substances

The prescribing of a medicinal product that is liable to abuse requires special attention and may be subject to specific statutory requirements. Practitioners may need to be authorized to prescribe controlled substances; in such cases it might be necessary to indicate details of authority on the prescription.

In particular, the strength, directions and the quantity of the controlled substance to be dispensed should be stated clearly, with all quantities written in words as well as in

figures to prevent alteration. Other details such as patient particulars and date should also be filled in carefully to avoid alteration.

References

1. *The Guide to Good Prescribing;* Geneva: WHO; 1994
2. *WH0 Model Formulary 2008.*
3. www.cdsco.nic.in/pharmacovigilance.htm

Chapter 2

COMMON CONDITIONS

FEVER

A body temperature greater than 37.7°C is defined as fever. Fever may be continuous, intermittent or remittent.

Investigations

A meticulous history of chronology of symptoms, any associated focal symptom(s), exposure to infectious agents and occupational history may be useful. A thorough physical examination repeated on a regular basis may provide potentially diagnostic clues.

- If there are no clinical clues, the tests include a complete haemogram with ESR, smear for malaria parasite, blood culture, Widal test and urine analysis including urine culture.
- If the febrile illness is prolonged to more than 2 weeks, an X-ray chest is indicated even in the absence of respiratory symptoms. Any abnormal fluid collection should be sampled.

Treatment

Routine use of antipyretics in low-grade fever is not justified. This may mask important clinical indications. However, in acute febrile illnesses suggestive of viral or bacterial cause, fever should be treated.

Non pharmacological treatment

Hydrotherapy and rest

Pharmacological treatment

Non-specific

- Tab. paracetamol 500-1000 mg 6-8 hourly. Or Tab. ibuprofen 400-600 mg 8 hourly.

Specific

- Antibiotics/antimalarials depending upon the cause suggested by clinical and laboratory evaluation.

Outcome

In most cases of fever, patient may either recover spontaneously or a diagnosis is reached after repeated clinical evaluation and investigations. If no diagnosis is reached in up to 3 weeks, patient is said to be having fever of unknown origin (FUO) and should be managed accordingly.

Patient education

- Self medication should be avoided.
- Antibiotics should be taken only on advice of a physician.
- Avoid covering the patient having high fever with blanket etc.
- Plenty of fluids should be taken. Stay in cool environment. Cold sponging of face and limbs should be done repeatedly.

References

1. Alterations in Body Temperature. In: Harrison's Principles of Internal Medicine. Kasper DL, Braunwald E, Fauci AS et al (eds), 16th Edition, 2005, McGraw Hill Company Inc., New York, pp 104-25.
2. Physiological Changes in Infected Patients. In: Oxford Textbook of Medicine. Warrell DA, Cox TM, Firth JD, Benz EJ Jr (eds), 4th Edition, 2003. Oxford University Press, 1289-92.

FEVER OF UNKNOWN ORIGIN (FUO)

FUO is defined as the presence of fever of 38.3°C or more recorded on several occasions, evolving for at least 3 weeks with no diagnosis reached even after one week of relevant and intelligent investigations. FUO is usually an uncommon presentation of common diseases.

Salient features

- Common causes include
- Infections, particularly extrapulmonary tuberculosis, typical and atypical mycobacterial infections, *Pneumocystis carinii* pneumonia, toxoplasmosis, lymphoma etc. In neutropenic patients, gram negative bacterial, staphylococcal and fungal infections are important causes.
- Neoplastic conditions
- Noninfectious inflammatory causes collagen vascular diseases, drug fevers
- Prolonged unexplained fever, often with no localizing clue on history, physical examination and basic laboratory investigations.

Investigations

- A detailed clinical history, repeated physical examination to provide potentially diagnostic clues (PDC).
- No single algorithmic approach to diagnosis can be recommended for all patients of FUO and diagnostic approach needs to be individualized.
- A complete haemogram including peripheral blood smear for malaria parasite, serum biochemistry particularly liver function tests, a tuberculin test and an X-ray of chest should be done in every patient with prolonged fever.
- Other investigations which are often helpful include tests related to collagen vascular diseases, an ultrasonography to localize intra-abdominal foci of infections and a contrast enhanced computed tomography (CECT) of chest and abdomen in detecting mediastinal lymph nodes and parenchymal lung abnormalities not seen on conventional chest X-ray.
- Further, diagnostic approach should take into consideration the PDCs from the evaluation of history, results of repeated physical examination, basic investigations and any investigations done prior to this episode. If any abnormal or doubtful lesion is detected FNAC/biopsy should be obtained.

Treatment

Sometimes evaluation may need discontinuation of all drugs being taken by the patient to rule out drug fever as the cause of FUO.

Symptomatic treatment for fever *(for details see section on fever)*

- Cold sponging may be done if fever produces discomfort. The emphasis in patients with classic FUO is on continued observation and examination.
- (Caution: Avoid 'shotgun' trials. Empirical therapies consisting of therapeutic trials commonly used in patients with FUO are: antibiotics, anti-tubercular treatment (ATT) and corticosteroids).
- *The ability of glucocorticoids and NSAIDs to mask fever while permitting the spread of infection dictates that their use should be avoided unless infection has been largely ruled out.*
- If on the basis of clinical evaluation and inability to reach a definitive diagnosis, a therapeutic trial is started, the following principles must be kept in mind:
  - Give only one set of trial at a given time
  - The doses of drugs and period of therapeutic trial must be adequate-
  - The patient must be followed closely for response

Patient education

- Self medication should be avoided.
- Antibiotics should be taken only on advice of a physician.
- Plenty of fluids should be taken. Stay in a cool environment.

ACUTE RHEUMATIC FEVER

Rheumatic fever is due to an autoimmune reaction to group streptococci and involves many organs, primarily the heart, the joints and the central nervous system.

Salient features

- Major criteria are pancarditis**,** polyarthritis**,** chorea**,** erythema marginatum and subcutaneous nodule.
- Minor criteria are arthralgia**,** fever**,** raised PR interval.

Non pharmacological treatment

- Bed rest for two weeks followed by gradual ambulation

Pharmacological treatment

- Tab. penicillin V 500 mg BD for 10 days or Inj. benzathine penicillin G- 1.2 million units in single does or Tab. erythromycin 250 mg QID for 10 days – if patient is allergic to penicillin.
- For secondary prophylaxis for rheumatic fever, give Inj. benzathine penicillin 1.2 million units every 3 weeks or Tab. penicillin V 250 mg BD daily or Tab. erythromycin 250 mg twice daily.
- Duration of prophylaxis is given in Table 1.

Table 1. Duration of secondary prophylaxis in acute rheumatic fever

Category of Patient	Duration of Prophylaxis
Rheumatic fever without carditis	For 5 years after the last attack or 21 years of age (whichever is longer).
Rheumatic fever with carditis but no residual valvular disease	For 10 years after the last attack, or 21 years of age (whichever is longer).
Rheumatic fever with persistent valvular disease, evident clinically or on echocardiography	For 10 years after the last attack, or 40 years of age (whichever is longer). Sometimes lifelong prophylaxis may be required.

- Arthritis**:** Tab. aspirin 2 g QID in adults and 80-100 mg/kg/day in children till symptoms persist or Tab. naproxen: 10-20 mg/kg/day can also be used.
- Carditis**:** Tab. prednisolone 1-2 mg/kg/day can be used for short periods or up to a maximum of 3 weeks.
- Chorea**:** It occurs as a long term sequelae (3-6 months). Carbamazepine or sodium valproate is preferred to haloperidol (0.05 mg/kg/day) to reduce the abnormal movements but response may take up to 2 weeks. Treatment should continue up to 1-2 weeks after symptoms subside.

ANAEMIA

Anaemia is defined as a low haemoglobin level (less than 13.5 g/dl in males and less than 13.5 gm/dl in females). The common causes of anaemia in India includes,reduced production due to deficiency of iron, folic acid, rarely vitamin B_{12}, or an ineffective erythropoiesis secondary to many causes (anaemia of chronic disease, secondary to infections and inflammation, endocrinal disorders, primary bone marrow disorders like infiltration or hypoplasia), blood loss (which also leads to iron deficiency) and increased destruction of RBCs (haemolysis due to many causes of which thalassaemia is the commonest).

Salient features

- Tiredness, weakness and lack of desire to work.
- Nails and tongue look pale. Severe anaemia produces general pallor and swelling of feet.
- Most aetiologies may be determined on the basis of MCV performed in an accurate cell counter as follows:
- Low MCV - Iron deficiency or haemoglobfnopathy like thalassaemia.
- High MCV - Folic acid or B_{12} deficiency. Less commonly alcohol intake, liver disease, haemolysis and hypothyroidism.
- Normal MCV - Anaemia of chronic disease, primary bone marrow disorders, renal failure, haemolysis.
- In case of associated leucocyte and platelet abnormalities or if anaemia does not respond to therapy in 4 weeks despite correcting the apparent cause, a bone marrow examination by aspiration/biopsy should be performed.

Pharmacological treatment

Iron deficiency anaemia

- Treat the underlying cause: menorrhagia in women, gastrointestinal blood loss in all age groups including hookworm infestation, dietary deficiency, rarely malabsorption.
- Tab. ferrous sulfate 200 mg 3 times a day.
- Reduce the dose as haemoglobin rises to over 10 g/dl. Once haemoglobin is normal, continue with 1 tablet daily for at least three months.
- Other preparations of iron are not superior, but they can be tried if patient does not find ferrous sulfate suitable. These include ferrous fumarate and ferrous gluconate. The rate of rise of haemoglobin should be 1 g/dl per week.
- If this does not occur, consider ongoing blood loss, noncompliance, and associated haemoglobinopathy like thalassaemia carrier status, malabsorption, or an incorrect diagnosis.
- Parenteral iron does not lead to a faster rise in haemoglobin. It is indicated in the following situations: (i) malabsorption of iron, (ii) intolerance of oral iron, (iii) in late pregnancy to ensure that foetal stores of iron are replenished rapidly, (iv) if ongoing blood loss exceeds the capacity to absorb oral iron (like in inoperable malignancy) and (v) in noncompliant patient. There is danger of anaphylactoid reactions; hence facilities to manage these should be readily available.

Folic acid deficiency

- Treat the cause: Dietary deficiency, increased requirement as in pregnancy and children, haemolytic anaemia.
- Tab. folic acid 5 mg daily. This dose is adequate even in malabsorption syndrome.

Vitamin B_{12} deficiency

- Treat the cause: Dietary deficiency in vegetarians, pernicious anaemia. Although uncommon, it is also under diagnosed due to lack of facilities.
- Tab. vitamin B complex, as in haematinic tablets. Or Inj. B_{12} 1000 mcg IM, one injection on alternate days for total 5 injections, then once a week for 5 weeks, then once in 3 to 6 months will be adequate for most patients.

Patient education

- Educate the patient about preventive measures for worm infestation.
- Inform about importance of taking adequate food with green leafy vegetables to meet the nutritional requirement.
- Cooking food in iron utensils may increase iron content in the diet.
- Instruct the patient to take iron tablets after food; iron tablets sometimes produce stomach upset.
- Inform the patient that the stools could turn black during iron therapy
- Reduce the dose of iron if it produces stomach ache, diarrhoea or constipation.
- Explain that the response to iron therapy is gradual and it takes weeks or months for blood to become normal. Continue iron tablets for six months.
- Keep iron tablets out of the reach of children. They may swallow the tablets as candies causing adverse reactions including death.

TYPHOID FEVER

It is caused by *Salmonella typhi* and *Salmonella paratyphi*. *Salmonella typhi* causes a variety of illnesses including asymptomatic carrier state, gastro-enteritis and enteric fever.

Salient features

- The onset of fever is typically gradual, continuous (temperature upto 40º C) constitutional symptoms such as malaise, anorexia, lethargy, headache.
- Constipation or diarrhea (pea-soup stools) which may be associated with abdominal pain and tenderness.
- Hepatomegaly and splenomegaly may be observed.
- Complications such as hepatitis, meningitis, pneumonitis and myocarditis can occur after the first week.

Investigations

- Diagnosis is suggested by
  - Rising of O antibodies (Widal test) and
  - Confirmed by isolation of organism in blood, bone marrow, urine or stool

Pharmacological treatment

- Most patients with uncomplicated enteric fever can be managed at home with oral antibiotics and antipyretics.

- Patients with persistent vomiting, diarrhea, and/or abdominal distension should be hospitalized and given supportive therapy as well as a parenteral third-generation cephalosporin or fluoroquinolone, depending on the susceptibility profile (Table 2).
- Therapy should be administered for at least 10 days or for 5 days after fever resolution.

Table 2. Dosage schedule of antibiotics in treatment of typhoid fever

Indication	Agent	Dosage (Route)	Duration, Days
Empirical Treatment	Ceftriaxone[a]	1-2 g/d IV	7–14
Fully Susceptible	Ciprofloxacin[b] (first line)	500 mg BD (PO) or 400 mg BD (IV)	5–7
	Amoxicillin (second line)	1 g TDS (PO) or 2 g QID (IV)	14
	Chloramphenicol	25 mg/kg TDS (PO or IV)	14–21
	Trimethoprim-sulfamethoxazole	160/800 mg BD (PO)	7–14
	Ciprofloxacin	500 mg BD (PO) or 400 mg BD (IV)	5–7
Multidrug-Resistant	Ceftriaxone	2–3 g/day (IV)	7–14
	Azithromycin	1 g/day (PO)[c]	5
	Ceftriaxone	2–3 g/d (IV)	7–14
	Azithromycin	1 g/d (PO)	5
Nalidixic Acid–Resistant	High-dose ciprofloxacin	750 mg bid (PO) or 400 mg q8h (IV)	10–14

[a]Or another third-generation cephalosporin [e.g., cefotaxime, 2 g TDS (IV); or cefixime, 400 mg bid (PO)].
[b]Or ofloxacin 400 mg bid (PO) for 2–5 days.
[c]Or 1 g on day 1 followed by 500 mg/day PO for 6 days.

- The 1–5% of patients who develop chronic carriage of *Salmonella* can be treated for 4–6 weeks with an appropriate oral antibiotic.
- Treatment with oral amoxicillin, cotrimoxazole, ciprofloxacin, or norfloxacin is 80% effective in eradicating chronic carriage of susceptible organisms.
- However, in cases of anatomic abnormality (e.g., biliary or kidney stones), eradication often requires both antibiotic therapy and surgical correction.

MALARIA

Parastitic infection due to protozoa of genus *Plasmodium* transmitted by the female anopheles mosquito. There are four plasmodia species: *Plasmodium falciparum, Plasmodium vivax, Plasmodium malaria, Plasmodium ovale.*

Salient features

- It is an acute and chronic illness characterized by paroxysms of fever, chills, sweats, fatigue, anemia and splenomegaly. *P. falciparum* malaria (severe and complicated malaria) is associated in varying degrees with the following clinical signs.
- Symptoms of cerebral malaria include mental clouding, coma, convulsions, delirium and occasionally localizing signs, hyperpyrexia (more than 40.5ºC), haemolysis, oliguria, anuria, pulmonary edema and macroscopic haemoglobinuria.

Investigations

Diagnosis is made by presence of protozoa in the blood in thick and thin smear slides (thick smear for easy detection of parasite and thin smear for identification of species).

Note: That blood films may be negative even in a severe attack because of parasites in the deep capillaries.

Pharmacological treatment (Figure 1, 2 and 3)

Figure 1. Treatment where microscopy result is available within 24 hours

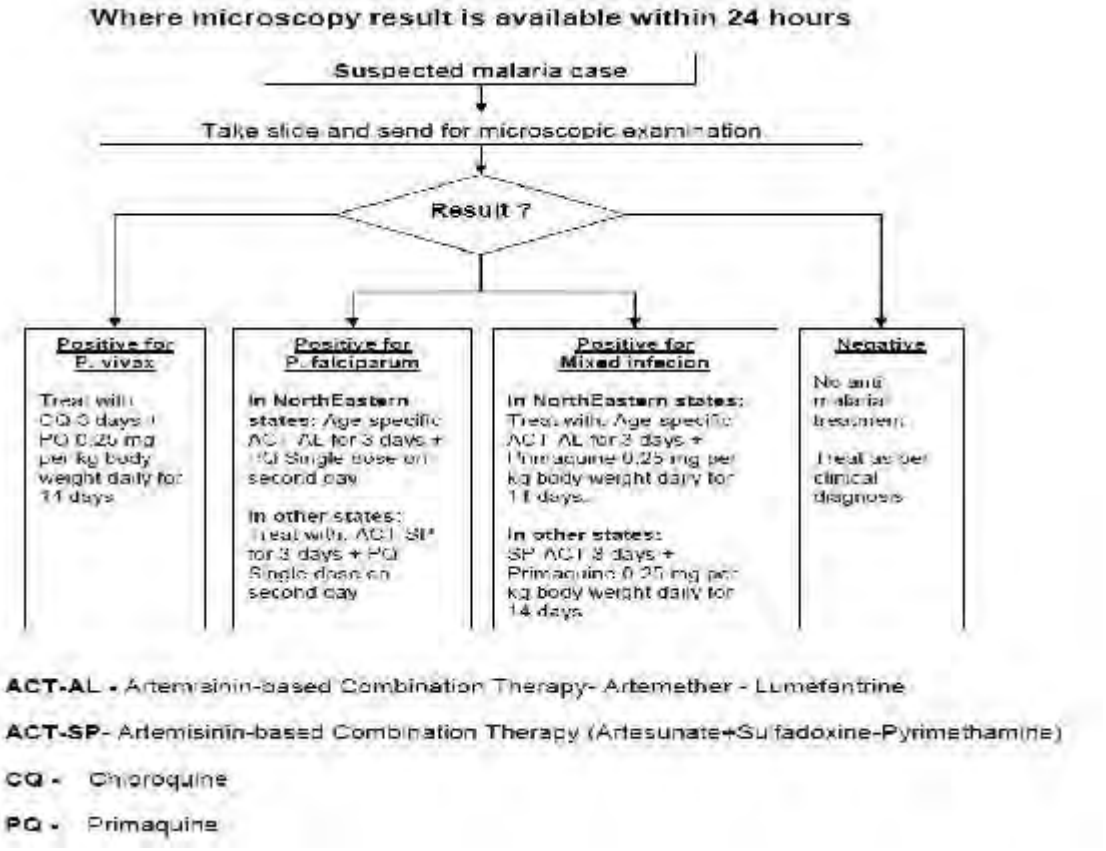
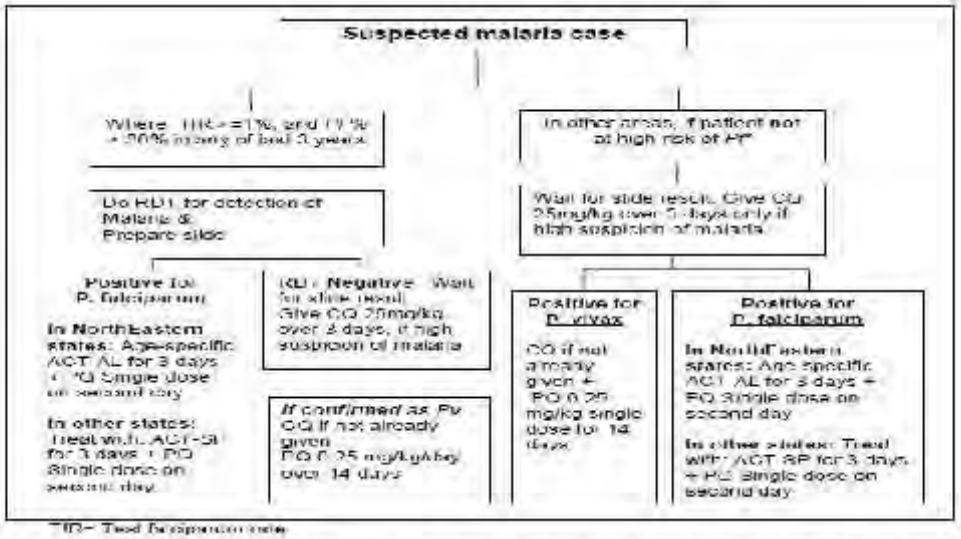


Figure 2. Treatment where microscopy result is not available within 24 hours and monovalent RDT is used



Note: If a patient has severe symptoms at any stage, then immediately refer to a nearest PHC or other health facility with indoor patient management or a registered medical doctor.

Note: PQ is contra-indicated in pregnancy and in children under 1 year (Infant).

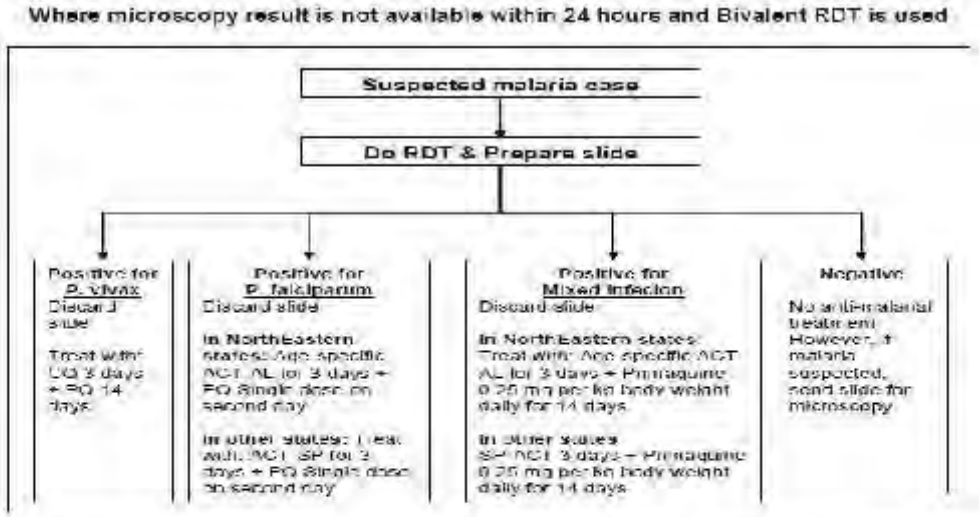
ACT-AL - Artemisinin-based Combination Therapy- Artemether – Lumefantrine

ACT-SP Artemisinin based Combination Therapy (Artesunate+Sulfadoxine Pyrimethamine)

CQ - Chloroquine

PQ - Primaquine

Figure 3. Treatment where microscopy result is not available within 24 hours and bivalent RDT is used



Note: if a patient has severe symptoms at any stage, then immediately refer to a nearest PHC or other health facility with indoor patient management or a registered medical doctor.

Note: PQ is contra-indicated in pregnancy and in children under 1 year (Infant).

ACT-AL - Artemisinin-based Combination Therapy- Artemether – Lumefantrine

ACT-SP Artemisinin based Combination Therapy (Artesunate+Sulfadoxine Pyrimethamine)

CQ - Chloroquine

PQ - Primaquine

Treatment of *P.vivax* malaria

- Tab. chloroquine: 25 mg/kg body weight divided over three days i.e.
  - 10 mg/kg on day 1
  - 10 mg/kg on day 2
  - 5 mg/kg on day 3
- Tab. primaquine: 0.25 mg/kg body weight daily for 14 days.
- Primaquine is contraindicated in infants, pregnant women and individuals with G6PD deficiency. 14 day regimen of primaquine should be given under supervision**.**

Treatment of *P. falciparum* malaria

- **Artemisinin based Combination Therapy (ACT-SP) –** Tab**.** artesunate 4 mg/kg body weight daily for 3 days plus sulfadoxine (25 mg/kg body weight) - pyrimethamine (1.25 mg/kg body weight) on first day.
- Tab. primaquine: 0.75 mg/kg body weight on day 2

Treatment of mixed infections *(P.vivax* + *P.falciparum)* cases

All mixed infections should be treated with full course of ACT and primaquine 0.25 mg per kg body weight daily for 14 days (Table 3).

Table 3. Dosage schedule for treatment of mixed infection *(P.vivax* + *P.falciparum)*

Age	Day 1			Day 2		Day 3		Days 4-14
	AS (50 mg)	SP (500 mg + 25 mg)	PQ (2.5 mg)	AS tablet (50 mg)	PQ (2.5 mg)	AS tablet (50 mg)	PQ (2.5 mg)	PQ (2.5 mg)
Less than 1 yr	1/2	1/2	0	1/2	0	1/2	0	0
1-4 years	1	1	1	1	1	1	1	1
5-8 years	2	1 .5	2	2	2	2	2	2
9-14 years	3	2	4	3	4	3	4	4
15 yrs or more	4	3	6	4	6	4	6	6

AS – artesunate, SP – sulfadoxine-pyrimethamine, PQ - primaquine

Treatment of complicated or severe malaria (Table 4)

Table 4. Treatment of severe and complicated malaria

Initial parenteral treatment for at least 48hrs. Choose any one of the following	**Follow-up treatment, when patient can take oral medications following parenteral treatment**
Quinine: 20mg quinine salt/kg body weight on admission (IV infusion or divided IM injection) followed by maintenance dose of 10 mg/kg 8 hourly; infusion rate should not exceed 5 mg/kg per hour. Loading dose of 20mg/kg should not be given, if the patient has already received quinine.	Quinine 10 mg/kg three times a day plus doxycycline 100 mg once a day or clindamycin in pregnant women and children under 8 years of age, to complete 7 days of treatment.
Artesunate: 2.4 mg/kg IV or IM given on admission, then at 12 h and 24 h, then once a day or **Artemether:** 3.2 mg/kg IM given on admission then 1.6 mg/kg per day or **Arteether:** 150 mg daily IM for 3 days in adults only (not recommended for children).	Treat with: ACT-SP for 3 days + PQ single dose on second day to complete 7 days of treatment.

Note: The parenteral treatment in severe malaria cases should be given for minimum of 24 hours once started (irrespective of the patient's ability to tolerate oral medication earlier than 24 hours).

Chemoprophylaxis of malaria

Short term chemoprophylaxis (up to 6 weeks)

Doxycycline: 100 mg once daily for adults and 1.5 mg/kg once daily for children (contraindicated in children below 8 years). The drug should be started 2 days before travel and continued for 4 weeks after leaving the malarious area.

Chemoprophylaxis for longer stay (more than 6 weeks)

Mefloquine: 250 mg weekly for adults and should be administered two weeks before, during and four weeks after exposure.

DENGUE

Dengue is the most important emerging tropical viral disease of human beings in the world today. All four dengue virus (Dengue 1, 2, 3 and 4) infections may be asymptomatic, may lead to dengue fever (DF), dengue hemorrhagic fever (DHF) or when associated with plasma leakage may lead to hypovolemic shock and dengue shock syndrome (DSS).

Salient features

- Dengue fever is an acute febrile illness of 2-7 days with two or more of the following manifestations. Headache, retro orbital pain, myalgia, arthralgia, rash.
- Hemorrhagic manifestation (petechiae and positive tourniquet test) and Leucopenia

Dengue hemorrhagic fever (DHF), if one or more of the following are present

- Positive tourniquet test
- Petechiae, purpura or ecchymosis
- Bleeding from mucosa
- Haematemesis, melena
- Thrombocytopenia (platelets 100,000 cells/mm^3 or less) and evidence of plasma leakage.

Dengue shock syndrome (DSS)

- All the above criteria of DHF plus signs of circulatory failure.

Notes: The tourniquet test is performed by inflating a blood pressure cuff to mid way between the systolic and diastolic pressure.

Non pharmacological treatment

- Bed rest is advisable during the acute phase.
- Use cold sponging to keep temperature below 39ºC

Pharmacological treatment

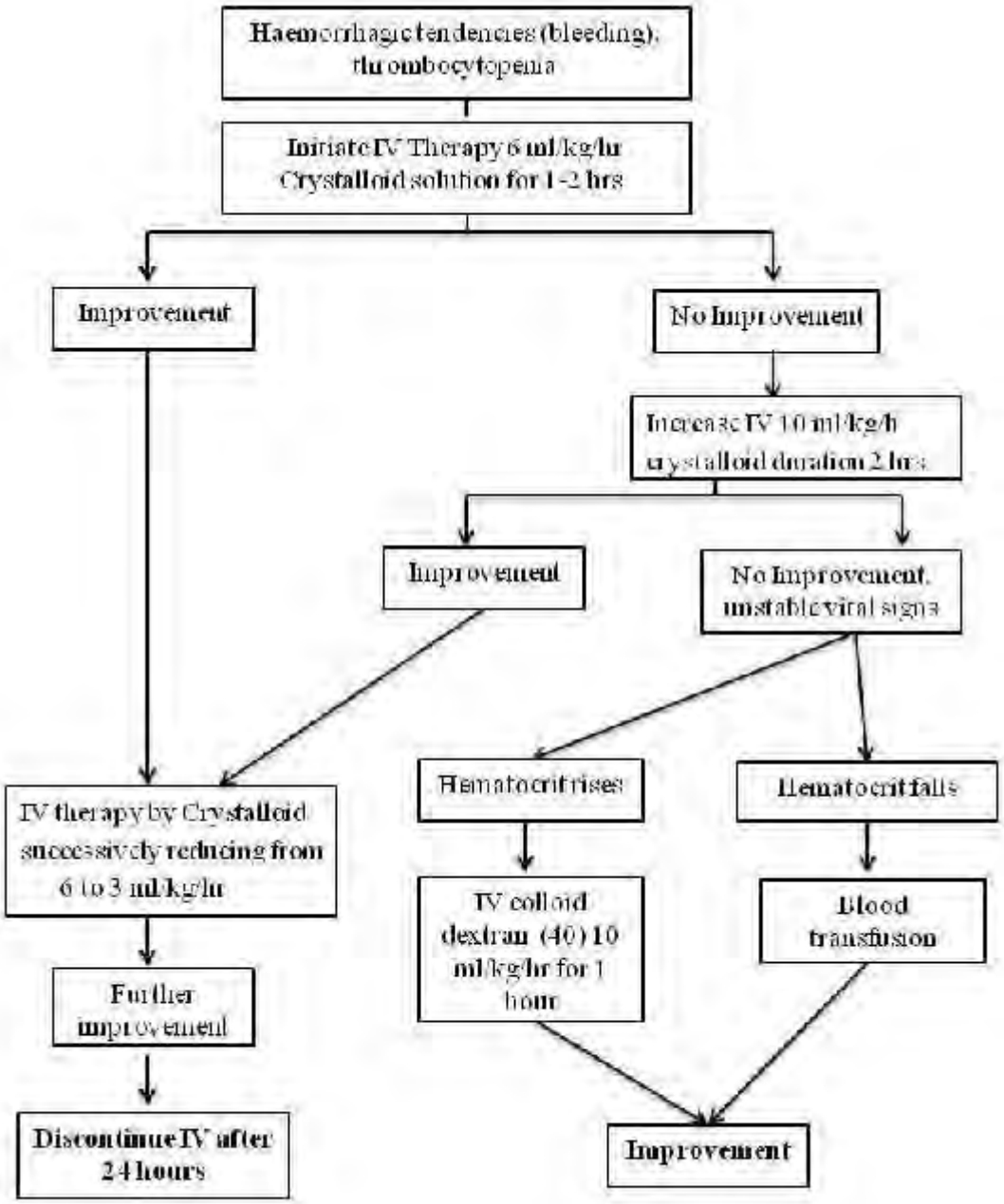
- Management of dengue fever is symptomatic and supportive
- Antipyretics may be used to lower the body temperature. Aspirin/NSAIDs like ibuprofen etc should be avoided since it may cause gastritis, vomiting, acidosis and platelet dysfunction.
- Paracetamol is preferable in the doses as follows: 1-2 years: 60 –120 mg/doses, 3-6 years: 120 mg/dose, 7-12 years: 240 mg/dose, Adult : 500mg/dose
  Note: In children the dose is calculated as per 10mg/kg body weight per dose which can be repeated at the interval of 6 hrs.
- Oral fluid and electrolyte therapy are recommended for patients with excessive sweating or vomiting.
- Patients should be monitored in DHF endemic area until they become afebrile for one day without the use of antipyretics and after platelet and haematocrit determinations are stable, platelet count is more than 50,000/mm^3.

Management of Dengue Hemorrhagic Fever (Febrile Phase)

- The management of febrile phase is similar to that of DF.
- Paracetamol is recommended to keep the temperature below 39°C. Copious amount of fluid should be given orally, to the extent the patient tolerates, oral hydration solution (ORS), such as those used for the treatment of diarrhoeal diseases and/or fruit juices are preferable to plain water

- IV fluid may be administered if the patient is vomiting persistently or refusing to feed.
- Patients should be closely monitored for the initial signs of shock. The critical period is during the transition from the febrile to the afebrile stage and usually occurs after the third day of illness.
- Serial haematocrit determinations are essential guide for treatment, since they reflect the degree of plasma leakage and need for intravenous administration of fluids.
- Haematocrit should be determined daily from the third day until the temperature has remained normal for one or two days. If haematocrit determination is not possible, haemoglobin determination may be carried out as an alternative.
- The details of IV treatment when required for patients are given in Figure 4.

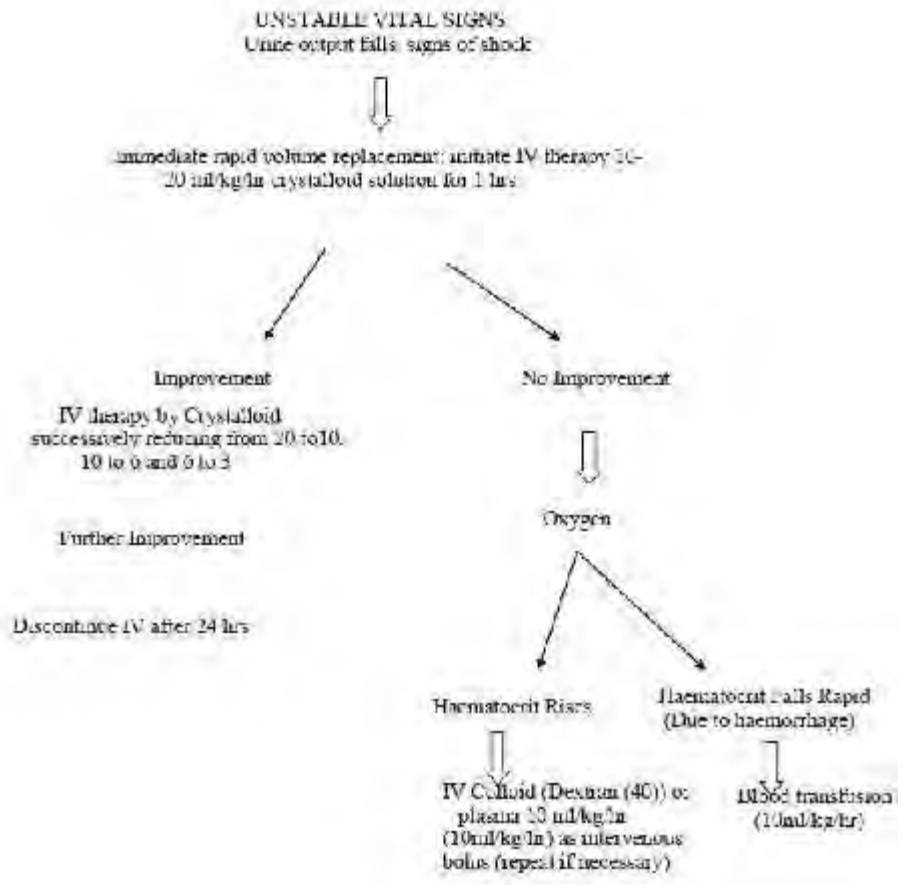
Figure 4. Treatment algorithm of dengue



Management of DHF Grade I and Grade II:

- Any person who has dengue fever with thrombocytopenia and haemoconcentration and presents with abdominal pain, black tarry stools, epistaxis, bleeding from the gums and infection etc needs to be hospitalized.
- All these patients should be observed for signs of shock. The critical period for development of shock is transition from febrile to abferile phase of illness, which usually occurs after third day of illness.
- A rise of haemoconcentration indicates need for IV fluid therapy. If despite the treatment, the patient develops fall in BP, decrease in urine output or other features of shock, the management for Grade III/IV DHF/DSS should be instituted.
- Oral rehydration should be given along with antipyretics like paracetamol sponging, etc. as described above.
- The detailed treatment for patient with DHF Grade I and II is given at Figure 5. Common signs of complications are observed during the afebrile phase of DHF. Immediately after hospitalization, the haematocrit, platelet count and vital signs should be examined to assess the patient's condition and intravenous fluid therapy should be started. The patient requires regular and sustained monitoring.

Figure 5. Volume replacement flow chart for patients with dengue haemorrhagic fever grade I and II



Fluid requirement

- The volume of fluid required to be replaced should be just sufficient to maintain effective circulation during the period of plasma leakage. To ensure adequate fluid replacement and avoid over-fluid infusion, the rate of intravenous fluid should be adjusted throughout the 24 to 48 hour period of plasma leakage by periodic haematocrit determinations and assessment of vital signs.
- The required regimen of fluid should be calculated on the basis of body weight and charted on a 1-3 hourly basis, or more frequently in the case of shock. The flow of fluid and the time of infusion are dependent on the severity of DHF. The schedule given below is a guideline and calculated for moderate dehydration of about 6% deficit (plus maintenance) (Table 5).

Table 5. Fluid requirement as per body weight of the patient

Weight on admission (kg)	Fluid requirement/kg body weight/day (ml/kg)
<7	220
7-11	165
12-18	130
>18	90

- In older children who weigh more than 40 kg, the volume needed for 24 hours should be calculated as twice that required for maintenance (using the Holiday and Segar formula). The maintenance fluid should be calculated as follows:

Table 6. Maintenance fluid requirement according to Holiday and Segar formula

Body weight in kg	Maintenance volume for 24 hours
<10kg	100 ml / kg
10-20kg	1000+50 ml / kg
>20kg	1500+20 ml / kg

For a child weighing 40 kgs, the maintenance is: 1500 + (20x20) = 1900 ml. This means that the child requires 3800 ml IV fluid during 24 hours.

Indications of red cell transfusion

- Loss of blood (overt blood) -10% or more of total blood volume
- Preferably whole blood/ component to be used
- Refractory shock despite adequate fluid administration and declining haematocrit - replacement volume should be 10 ml/kg body wt at a time and coagulogram should be done.

- If fluid overload is present, PCV is to be given

Indications of platelet transfusion

- In general there is no need to give prophylactic platelets even at less than 20,000 cells/mm^3.
- Prophylactic platelet transfusion may be given at level of less than 10,000 cells/mm^3 in absence of bleeding manifestations.
- Prolonged shock; with coagulopathy and abnormal coagulogram.
- In case of systemic massive bleeding, platelet transfusion may be needed in addition to red cell transfusion.

CHIKUNGUNYA

Chikungunya is caused by an alpha virus closely related to O' nyong – nyong virus. The main vector is Aedes aegypti mosquito.

Salient features

- Acute self limiting illness Incubation period is of 2 to 4 days. Abrupt onset presenting as fever with severe joint pain
- After 1 – 4 days, fever subsides, there will be a afebrile period 3 days, fever returns with an itching maculopapular rash on trunk and extensor surfaces of limbs.
- After another 3-6 days fever subsides and there is complete recovery.
- Cripping arthropathy can occur intermittently for upto 4 months, in some cases even upto 5 years.

Non pharmacological treatment

- Mild exercises and physiotherapy may be suggested in recovering persons.
- Exposure to warm environment may be suggested
- Non weight bearing exercises
- Surgery in severely damaged joints

Pharmacological treatment

- Acute stage of illness
- Treat symptomatically (Tab. paracetamol 1 gm 3-4 times a day for fever, headache and pain.)
- Avoid aspirin and steroid because of risk of GI side effects and Reye's syndrome with aspirin.
- Tab. hydroxychloroquinine 200mg orally once daily or Tab. chloroquinine phosphate 300mg per day for a period of 4 weeks in cases where arthralgia is refractory to other drugs.
- Side effects of hydroxychloroquinine include retinal damage and elevated liver enzymes.

TUBERCULOSIS

Salient features

Pulmonary tuberculosis (PTB): The most common symptom of PTB is a persistent cough of two weeks or more, with or without expectoration. It may be accompanied by one or more of the following symptoms,

- Fever, night sweats, weight loss
- Chest pain, hemoptysis, shortness of breath, tiredness and loss of appetite

Such patients should be selected and subjected for sputum examination. This enhances the chances of detection of the bacilli in the smear microscopy.

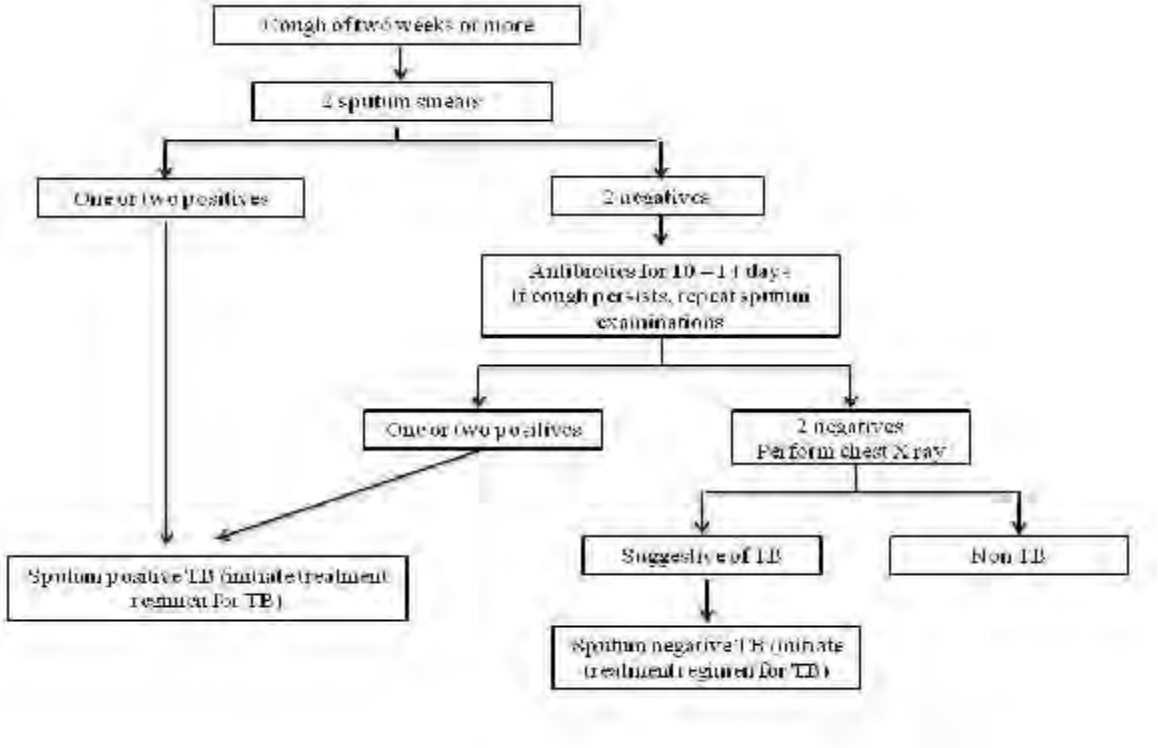
Extra pulmonary tuberculosis: A person with extra-pulmonary TB may have symptoms related to the organs affected along with constitutional symptoms stated above.

- Enlarged cervical lymph nodes with or without discharging sinuses (TB Lymphadenitis)
- Chest pain with or without dyspnoea in pleural TB
- Pain and swelling of the joints in bone tuberculosis (fever, backache, deformity in spinal TB).
- Signs of raised intra-cranial tension like irritability, headache, vomiting, fever, stiffness of the neck and mental confusion in TB meningitis
- Painless haematuria or sterile pyuria in renal tuberculosis and, infertility in genito-urinary TB.

Investigations

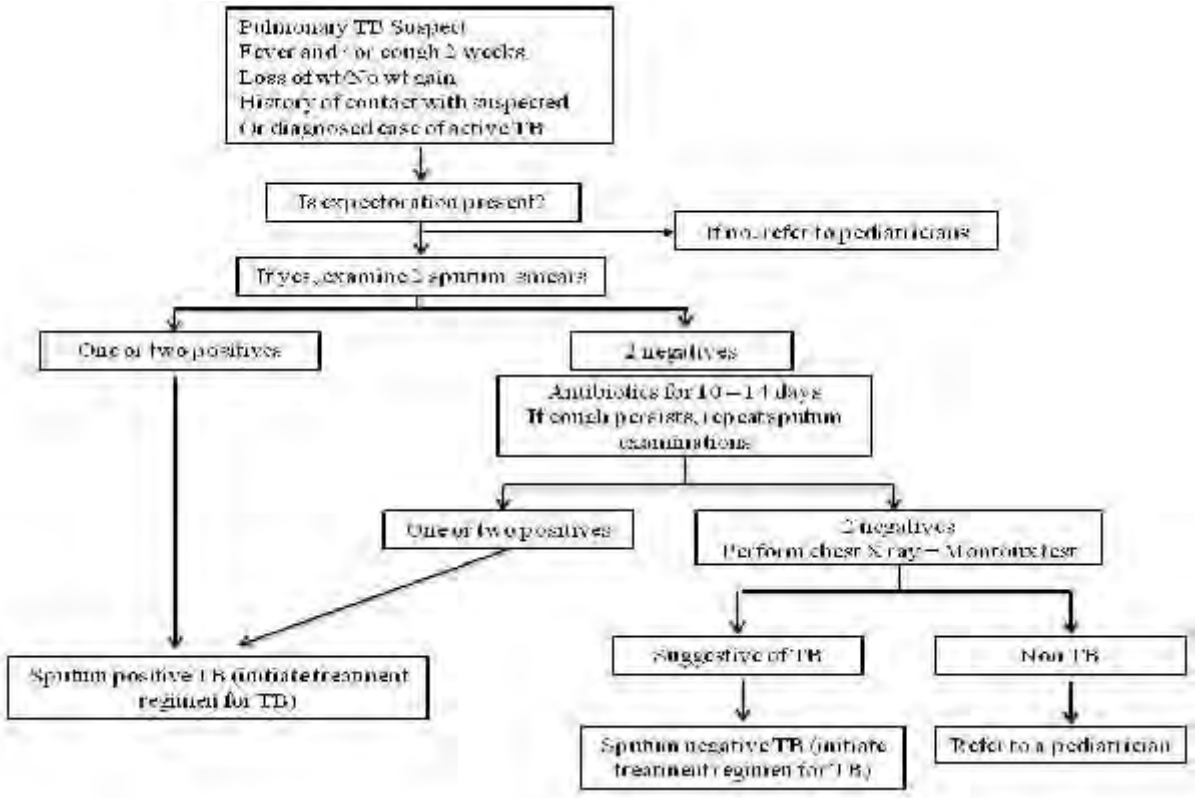
- **Pulmonary TB in adults (Figure 6)**
  - Sputum smear microscopy
  - Chest x-ray
  - Sputum culture and drug sensitivity testing(DST) for diagnosis of Drug Resistant TB.
  - Newer rapid diagnostic tools for detection of multi drug resistant (MDR) TB.
  - Newer tools under evaluation for diagnosis of MDR/XDR TB

Figure 6. Algorithm for diagnosis of pulmonary tuberculosis



- **Paediatric TB patients (Figure 7)**
  - In all children with presumptive intra-thoracic TB, microbiological confirmation should be sought through examination of respiratory specimens (like sputum by expectoration, gastric aspirate, induced sputum, bronchio-alveolar lavage or any other appropriate specimen) with diagnostic test like smear microscopy or rapid molecular test or culture.
  - In the event of negative or unavailable microbiological results, a diagnosis of probable TB in children should be based on the presence of abnormalities consistent with TB on radiography, a history of exposure to an infectious case, evidence of TB infection and clinical findings suggestive of TB.
  - For children with presumptive extra-pulmonary TB, appropriate specimens from the presumed sites of involvement should be obtained for microscopy, culture and DST, rapid molecular test and histopathological examination.

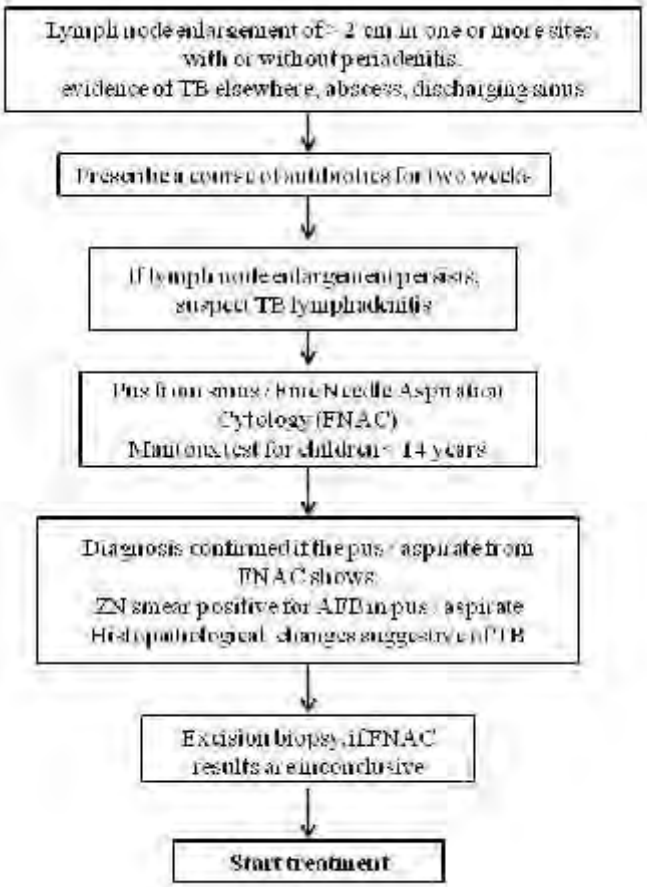
Figure 7. Diagnostic algorithm for pediatric pulmonary tuberculosis



- **Extra-pulmonary TB**
  - Demonstration of AFB bacilli in a smear from extra pulmonary site is often difficult because of low bacillary load. The clinical features pertaining to the system affected should be considered in the diagnosis of extra pulmonary tuberculosis.
  - However, the following are some of the special investigations which are helpful in diagnosing extra pulmonary tuberculosis. These may be radiological, cytological / pathological, biochemical and immunological (Figure 8).

(a) Fine Needle Aspirations Cytology and direct smear examination
(b) Excision / Biopsy of specimen for histopathological examination
(c) Fluid for cytology , biochemical analysis and smear examination
(d) X-ray of the involved region
(e) Ultrasonography for abdominal tuberculosis
(f) Culture for *Mycobacterium tuberculosis*
(g) Computerized tomography (CT) and magnetic resonance imaging (MRI)

Figure 8. Diagnostic algorithm for TB lymphadenitis



- **Diagnosis of Pleural TB**
  - Tubercular pleural effusion is considered as extra pulmonary tuberculosis. TB Patients presenting with chest pain with or without difficulty in breathing for more than two weeks should be referred for chest radiograph.
  - In pleural effusion, chest X-ray features may include obliteration of costophrenic angle and varying degree of effusion.
  - Biochemical and cytological analysis of the aspirated pleura fluid will help in confirming the diagnosis. Pleural fluid is generally an exudate with mainly lymphocytes and few mesothelial cells.
  - Pleural biopsy is confirmatory in a high proportion of patients.

Pharmacological treatment

Revised National Tuberculosis Control Programme (RNTCP) uses *short course chemotherapy* given intermittently - thrice weekly under *Direct Observation* for both pulmonary and extra pulmonary tuberculosis patients.

Treatment regimens

For the purpose of treatment regimen to be used, TB patients are classified into two groups, namely, “New” or “Previously Treated”, based on the history of previous treatment.

- **Regimen for New cases**: This regimen is prescribed to all new pulmonary (smear positive and negative) and extra pulmonary tuberculosis. The regimen is ***$2H_3R_3Z_3$ E_3 / 4 H_3R_3.***
- **Regimen for Previously Treated cases**: This regimen is prescribed for TB patients who have had more than one month anti-tuberculosis treatment previously. These patients are at a higher risk of having drug resistance. Hence, 5 drugs are prescribed in the intensive phase and the total duration of treatment is 8 months. Relapses, treatment after Default, Failures and Others are treated with this regimen. The regimen is *$2S_3H_3R_3Z_3$ E_3 / $1H_3R_3Z_3$ E_3 / 5 H_3R_3 E_3*. The table below indicates the treatment regimen, type of patients and regimens prescribed (Table 6).

Table 6. Treatment regimens recommended by RNTCP for treatment of Tuberculosis

Treatment groups	Type of patient	Regimen1	
		Intensive Phase (IP)	**Continuation Phase (CP)**
New*	Sputum smear-positive Sputum smear-negative Extra-pulmonary Others	$2H_3R_3Z_3E_3$	$4H_3R_3$
Previously Treated**	Smear-positive relapse Smear-positive failure Smear-positive treatment after default Others[1]	$2H_3R_3Z_3E_3S_3$ / $1H_3R_3Z_3E_3$	$5H_3R_3E_3$

* New includes former categories I and III

** Previously treated is former category II.

1 - In rare and exceptional cases, patients who are sputum smear-negative or who have extra-pulmonary disease can have recurrence or non-response. This diagnosis in all such cases should always be made by an MO and should be supported by culture or histological evidence of current, active TB. In these cases, the patient should be typed as 'Others' and given treatment regimen for previously treated

- The number before the letters refers to the number of months of treatment. The subscript after the letters refers to the number of doses per week.
- The dosage strengths are as follows: isoniazid (H) 600 mg, rifampicin (R) 450 mg, pyrazinamide (Z) 1500 mg, ethambutol (E) 1200 mg, streptomycin (S) 750 mg.
- Patients who weigh 60 kg or more receive additional rifampicin 150 mg.
- Patients more than 50 years old receive streptomycin 500 mg. Patients who weigh less than 30kg, receive drugs as per paediatric weight band boxes according to body weight (Table 7).

Table 7. Drug dosages for adults in the blister packs

Drugs	Dose (thrice a week)	Number of tablets in blister pack
Isoniazid (H)	600mg	2 x 300 mg
Rifampicin (R)	450mg	1 x 450 mg
Pyrazinamide (Z)	1500mg	2 x 750 mg
Ethambutol (E)	1200mg	2 x 600 mg
Streptomycin (S)	0.75g	-

- **Treatment of Pediatric TB (Table 8, 9 and 10, Figure 9)**
  Pediatric cases are to be treated under RNTCP with the same thrice weekly short course chemotherapy regimens ("New" or "Previously treated") given under Directly Observed Treatment Short Course (DOTS) as for adult patients. They are to be registered in the respective RNTCP TB Register. Pediatric patient-wise boxes are available with different dosages under two weight bands for children weighing 6 to 10 kg and 11 to 17 kg. Wherever possible, before a child is started on the "Previously Treated" regimen, she/he should be examined by a pediatrician or TB expert.

Table 8. Pediatric patient wise boxes (PWB) with dosages

DRUGS	Product code (PC) -13		Product code (PC) -14		PC-15	PC-16
	IP 24 blisters	CP 18 blisters	IP 24 blisters	CP 18 blister	12 blisters	12 blisters
Isoniazid	75 mg	75 mg	150 mg	150 mg	Prolongation of IP for PC 13	Prolongation of IP for PC 14
Rifampicin	75 mg	75 mg	150 mg	150 mg		
Pyrazinamide	250 mg		500 mg			
Ethambutol	200 mg		400 mg			

IP – Intensive phase, CP – Continuation phase

Table 9. Pediatric patient wise boxes (PWB) for new cases according to weight band

Weight band	For New cases		Prolongation of IP
	IP	CP	
6-10 Kg	PC 13		PC 15
11-17 Kg	PC 14		PC 16
18-25 Kg	PC 13 + PC 14		PC 15 + PC16
26-30 kg*	PC 14 x 2		PC 16 x 2

* For children weighing >30 kgs, adult PWB are to be used

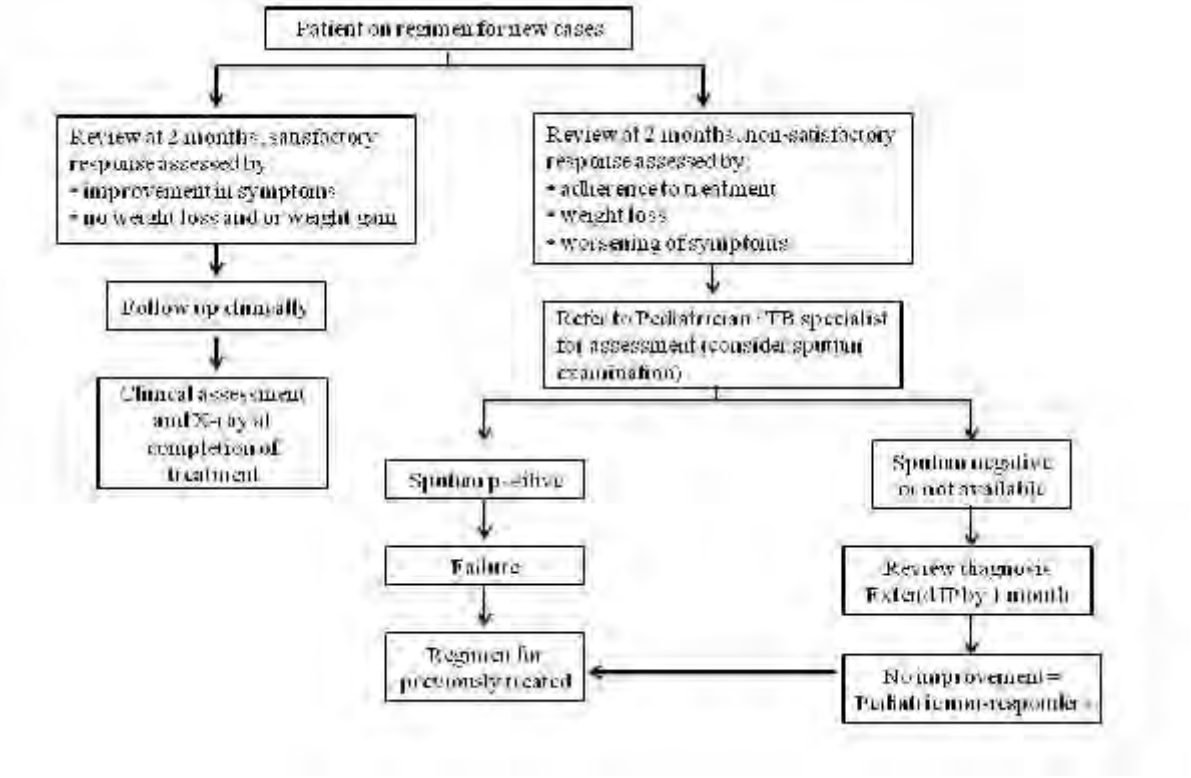
IP – Intensive phase, CP – Continuation phase

Table 10. Pediatric patient wise boxes for previously treated cases according to weight band

Weight	For previously treated cases		Prolongation of IP
	IP (Intensive phase)	CP (Continuation phase)	
6-10 Kg	(PC 13+ PC15) + (24 vials Inj sm*)	(PC 13 + 54 Tab E 200 mg) + (PC 15 without Z)	PC 15
11-17 Kg	(PC 14 + PC16) + (24 vials Inj sm*)	(PC 14 + 54 E 400 mg) + (PC 16 without Z)	PC 16
18-25 Kg	(PC 13 + PC 14 + PC 15 + PC 16) + (24 vials Inj sm*)	(PC 13 + PC14 + 54 Tab E 600 mg) + (PC 15+ PC 16 without Z)	PC 15 + PC 16
26-30 kg	(PC 14 x 2)+ (PC 16 x 2) + (24 vials Inj sm*)	(PC 14 x 2+ 54 Tab E 800 mg) + (PC 16 x 2 without Z)	PC 16 x 2

* Injection streptomycin 15mg/kg body weight

Figure 9. Clinical monitoring of children with Tuberculosis



Non DOTS (ND) treatment regimen under RNTCP

In rare and exceptional situations, non-DOTS treatment (with a self-administered non-rifampicin containing regimen) may be needed in a few TB cases. Examples include:

- Those with adverse reactions to rifampicin and / or pyrazinamide
- "New" patients who refuse DOTS despite all efforts

This is a treatment regimen of 12-month duration comprising 2 months of SHE and 10 months of HE (2SHE / 10HE). Dosages administered per day in the regimen are:

Isoniazid	-	300 mg
Ethambutol	-	800 mg
Streptomycin	-	750 mg (500 mg for those >50 years of age)

Those who weigh less than 30 kg receive dosages calculated as per body-weight.

- **Drug resistant tuberculosis (MDR/XDR-TB)**

MDR TB suspect: A patient suspected of drug-resistant tuberculosis, based on RNTCP criteria for submission of specimens for drug-susceptibility testing.

MDR TB suspect criteria

Criteria A

- All failures of new TB cases
- Smear positive previously treated cases who remain smear positive at 4th month onwards
- All pulmonary TB cases who are contacts of known MDR TB case

Criteria B (in addition to Criteria A)

- All smear positive previously treated pulmonary TB cases at diagnosis
- Any smear positive follow up result in new or previously treated cases

Criteria C (in addition to Criteria B)

- All smear negative previously treated pulmonary TB cases at diagnosis,
- HIV TB co-infected cases at diagnosis

In other words, for districts implementing MDR Suspect Criteria B, any smear-positive diagnostic (except in a new patient) or any smear-positive follow-up result, should prompt a referral for DST.

MDR-TB case: A TB patient whose sputum is culture positive for *Mycobacterium tuberculosis* and is resistant *in-vitro* to isoniazid and rifampicin with or without other anti-tubercular drugs based on DST results from an RNTCP-certified culture and DST laboratory

Extensive Drug Resistant Tuberculosis (XDR TB) case: An MDR TB case whose recovered *M. tuberculosis* isolate is resistant to at least isoniazid, rifampicin, a fluoroquinolone (ofloxacin, levofloxacin, or moxifloxacin) and a second-line injectable anti-TB drug (kanamycin, amikacin, or capreomycin) at a RNTCP-certified culture and DST laboratory. A patient is confirmed to have MDR or XDR TB only when the results are from a RNTCP quality-assured culture and DST Laboratory and by a RNTCP-endorsed testing method mentioned as below:

Choice of Diagnostic Technology

Currently, the programme is scaling up the laboratory capacity of various culture and DST laboratories at state level IRLs and other laboratories. The choice of technology to be used for diagnosis of MDR TB has been determined as per recommendations of the National Laboratory Committee. Thus, for DST at certified laboratory, wherever available Molecular DST (e.g. Line Probe Assay (LPA)) is preferred diagnostic method because of the rapid and highly-accurate rifampicin results, followed in preference by Liquid C-DST and then Solid C-DST.

MDR Diagnostic Technology Choice

First	Molecular DST (e.g. LPA DST)
Second	Liquid culture isolation and LPA DST
Third	Solid culture isolation and LPA DST
Fourth	Liquid culture isolation and Liquid DST
Fifth	Solid culture isolation and DST

Similarly for follow up cultures, wherever available, liquid culture will be preferred over solid culture. However, this will be liquid cultures for at least the crucial months of follow-up (IP-3, 4, 5, 6 and CP-18, 21, 24) and over and beyond this, it will be determined by the workload of individual laboratories.

Table 11. Grouping anti-TB agents (first line and second line drugs)

Grouping	Drugs
Group 1: First-line oral anti-TB agents	Isoniazid (H); Rifampicin (R); Ethambutol (E); Pyrazinamide(Z)
Group 2: Injectable anti-TB agents	Streptomycin (S); Kanamycin (Km); Amikacin (Am); Capreomycin (Cm); Viomycin (Vm).
Group 3: Fluoroquinolones	Ciprofloxacin (Cfx); Ofloxacin (Ofx); Levofloxacin (Lvx); Moxifloxacin (Mfx); Gatifloxacin (Gfx)
Group 4: Oral second-line anti-TB agents	Ethionamide (Eto); Prothionamide (Pto); Cycloserine (Cs); Terizadone (Trd); para-aminosalicylic acid (PAS)
Group 5: Agents with unclear efficacy (not recommended by WHO for routine use in MDR-TB patients)	Clofazimine (Cfz); Linezolid (Lzd); Amoxicillin / Clavulanate (Amx/Clv); thioacetazone (Thz); imipenem/cilastatin (Ipm / Cln); high-dose isoniazid (high-dose H); Clarithromycin (Clr)

- All drugs should be given in a single daily dosage under directly observed treatment (DOT) by a DOT Provider. All patients will receive drugs under direct observation on 6 days of the

week. On Sunday, the oral drugs will be administered unsupervised whereas injection kanamycin will be omitted.

- If intolerance occurs to the drugs, ethionamide, cycloserine and PAS may be split into two dosages and the morning dose administered under DOT. The evening dose will be self-administered. The empty blister packs of the self-administered doses will be checked the next morning during DOT.
- Pyridoxine should be administered to all patients on an RNTCP Category IV regimen.

Drug dosages for MDR-TB cases are base on the weight bands as shown in below (Table 12).

- **Treatment of MDR TB (Category IV regimen)**
  - RNTCP will be using a standardised treatment tegimen (Cat IV) for the treatment of MDR-TB cases (and those with rifampicin resistance) under the programme.
  - Cat IV regimen comprises of 6 drugs (Table 11) kanamycin, high dose levofloxacin, ethionamide, pyrazinamide, ethambutol and cycloserine during 6-9 months of the intensive phase (IP) and 4 drugs- high dose levofloxacin, ethionamide, ethambutol and cycloserine during the 18 months of the continuation phase (CP).
  - P-aminosalicylic acid (PAS) is included in the regimen as a substitute drug in case of intolerance i.e. severe ADR leading to discontinuation of drug to any of the drug in the Cat IV regimen.
  - For management of patients with Km mono resistance, the committee recommended replacement of Km with capreomycin, If Km is intolerant, substitute amikacin (AK) if possible, or PAS if injectable agent not feasible.
- **Drug dosages and administration**

Table 12. Dosage and weight band recommendations of second line drugs

S.No	Drugs	16-25 Kgs	26-45 Kgs	46-70 Kgs
1	Kanamycin	500 mg	500 mg	750 mg
2	Levofloxacin (High Dose)	250 mg	750 mg	1000 mg
3	Ethionamide	375 mg	500 mg	750 mg
4	Ethambutol	400 mg	800 mg	1200 mg
5	Pyrazinamide	500 mg	1250 mg	1500 mg
6	Cycloserine	250 mg	500 mg	750 mg
7	PAS (80% Bioavailability)[1]	5 gm	10 gm	12 gm
8	Pyridoxine	50 mg	100mg	100mg

[1] *In case of PAS with 60% bioavailability the dose will be increased to 7 gm (16-25 Kg); 14 gm (26-45 Kg) and 16 gm (> 45 Kg)*

- If a patient gains 5 kg or more in weight during treatment and crosses the weight-band range, the DOTS–Plus site committee may consider moving the patient to the higher weight-band drug dosages.

- Similarly if a patient loses 5 kgs or more in weight during treatment and crosses the weight band the DOTS Plus site committee may consider moving the patient to the lower weight band.
- The new higher/lower dosages are provided whenever the patient is due for the next supply of drugs in the normal course of treatment and not as soon as change of weight is noted.
- The dosages of 2nd line drugs for MDR TB cases in paediatric age group weighing < 16 kg as per the Table 13.

Table 13. Dosage of 2nd line drugs for treatment of pediatric MDR TB cases

Drug	Daily Dose – mg/kg body weight
Kanamycin	15-30
Levofloxacin	7.5-10
Ethionamide	15-20
Cycloserine	15-20
PAS	150
Ethambutol	25
Pyrazinamide	30-40

- The dosages for higher weight patients include use additional dosages of some 2nd line drugs for MDR TB cases in patients weighing > 70 kg taking the dosage to kanamycin (1 gm), ethionamide (1 gm), cycloserine (1 gm), ethambutol (1.6 gm) and pyrazinamide (2 gm). All these are well within the maximum permissible dosage for each drug as per the WHO guidelines.

- **Treatment of XDR-TB (Category V)**

All XDR-TB patients who are considered for category V regimen, will be screened by the DOTS-Plus committee and if found suitable will be referred to a thoracic surgeon for consideration of surgery to improve the treatment outcomes. Identification must be done for the site (tertiary centres) with such surgical facilities.

XDR Regimen (Table 14)

6-12 months Cm, PAS, Mfx, High dose-H, Cfz, Lzd, Amx/Clv

18 months PAS, Mfx, High dose-H, Cfz, Lzd, Amx/Clv

- The Category V treatment regimen consists of 7 drugs, with 2 reserve/substitute drugs. The dosage of the drugs would vary as per the weight of the patient (≤ 45Kg or > 45Kg). All drugs are to be given on a daily basis. Injections of capreomycin will be given for 6 days/week (not on Sundays).

- All morning doses are to be supervised by the DOT Provider except on Sundays. After taking DOT for morning doses on Saturday, next day medicines would be given to the patient to be taken at home on Sunday.
- Empty blisters of medicines taken unsupervised in evening and on Sundays are to be collected by DOT Provider.
- The change from IP to CP will be done only after achievement of culture conversion i.e. 2 consecutive negative cultures taken at least one month apart. In case of delay in culture conversion, the IP can be extended from 6 months up to a maximum of 12 months. In case of extension, the DOTS – Plus site Committee, which will be responsible for initiating and monitoring the Category V regimen, can decide on administering capreomycin injection intermittently (3 times/week) for the months 7 to 12.

Table 14. Drugs and Dosages for management of XDR TB

Drugs	**Dosage/day**	
	≤ 45 Kgs	**> 45 Kgs**
Inj. Capreomycin (Cm)	750 mg	1000 mg
PAS	10 gm	12 gm
Moxifloxacin (Mfx)	400 mg	400 mg
High dose INH (High dose-H)	600 mg	900 mg
Clofazimine (Cfz)	200 mg	200 mg
Linezolid (Lzd)	600 mg	600 mg
Amoxiclav (Amx/Clv)	875/125 mg BD	875/125 mg BD
Pyridoxine	100 mg	100 mg
Reserve/Substitute drugs		
Clarithromycin (Clr)	500 mg BD	500 mg BD
Thiacetazone (Thz)#	150 mg	150 mg

Depending on availability, not to be given to HIV positive cases

The reserve/substitute drugs would be used in the following conditions:

- In case the patient was on PAS in Category IV, PAS will be replaced with one of the reserve drugs in the Category V regimen
- If the patient is unable to tolerate one or more of the drugs
- If the patient is found to be resistant to capreomycin

- **HIV co-infection with TB**
  - TB patients living with HIV should receive the same duration of TB treatment with daily regimen as HIV negative TB patients.
  - Antiretroviral therapy (ART) must be offered to all patients with HIV and TB as well as drug-resistant TB requiring second-line anti-TB drugs, irrespective of CD4 cell-count, as early as possible (within the first 8 weeks) following initiation of anti-TB treatment. Appropriate arrangements for access to antiretroviral drugs should be made

for patients. However, initiation of treatment for TB should not be delayed. Patients with TB and HIV infection should also receive co-trimoxazole as prophylaxis for other infections.

- People living with HIV should be screened for TB using four symptom complex (current cough or, fever or weight loss or night sweats) at HIV care settings and those with any of these symptoms should be evaluated for ruling out active TB. Such patients in whom active TB is ruled out, Isoniazid Preventive Therapy (IPT) in the dosage of 10 mg/Kg body weight should be offered to them for 6 months.

- **Adherence**
  - Supervision and support should be individualized.
  - Ensuring Adherence in M/XDR TB patients: Patient support systems, including direct observation of treatment, are required to ensure adherence. It should be ensured that the patient consumes all the dosages of the drugs.
  - Trained treatment supporter for treatment adherence: Such measures may include identification and training of a treatment supporter (for TB and, if appropriate, for HIV, diabetes mellitus etc.) who is acceptable, accessible and accountable to the patient and to the health system.
  - Information Communication Technology to promote treatment literacy and adherence: Optimal use of Information Communication Technology (ICT) should be done to promote treatment literacy and adherence

Other important points to remember

- Public Health Responsibility- Any practitioner treating a TB patient is assuming an important public health responsibility to prevent ongoing transmission of the infection and the development of drug resistance. To fulfill this responsibility the practitioner must not only prescribe an appropriate regimen, but also utilize local public health services / community health services, and other agencies including NGOs when necessary, to assess the adherence of the patient and to address poor adherence when it occurs.
- Maintain Records for all TB patients: A written record of all medications given, microbiological response and adverse reactions should be maintained for all patients. Every patient's clinical outcome should be recorded in the case register and available for public health monitoring.
- Notification of TB Cases: All health establishments must report all TB cases and their treatment outcomes to public health authorities (District Nodal Officer for Notification).
  Proper feedback need to be ensured to all health care providers who refer cases to public health system on the outcome of the patients which they had referred.
- Contact Investigation:
  All providers of care for patients with TB should ensure all household contacts (including children) and other persons who are in close contact with TB patients are screened for TB as per defined diagnostic standards.

In case of pediatric TB patients, reverse contact tracing for search of any active TB case in the household of the child must be undertaken.

The highest priority contacts of known TB cases for intensive case finding (such as are:

- Persons with symptoms suggestive of TB
- Children aged <6 years
- Known or suspected immune-compromised patient, particularly HIV infection.
- Diabetes Mellitus
- Other higher risks including pregnancy, mental disorders, smokers and alcoholics etc.
- Patients with DR-TB - In case of contact with a DR-TB index case, close clinical monitoring should be provided, as there is no evidence that treatment of latent infection with available drugs is presently effective

Follow up

Follow up smear and culture schedule during treatment

- For follow up examination two sputum specimens will be collected and examined by smear and culture at least 30 days apart from the 3rd to 7th month of treatment (i.e. at the end of the months 3, 4, 5, 6 and 7) and at 3-monthly intervals from the 9th month onwards till the completion of treatment (i.e. at the end of the months 9, 12, 15, 18, 21 and 24).
- The two specimens for smear and culture at the RNTCP accredited lab (one early morning and one supervised spot) will be collected and transported from the respective DTC to the RNTCP accredited laboratory.
- If any of the cultures in the last three quarters becomes positive, it will be followed up by monthly cultures in the following 3 months.

Table 15. Common adverse effects of 2nd line drugs and management strategies

Adverse effects	Suspected agent(s)	Suggested management strategies
Gastrointestinal		
Nausea and vomiting	PAS, isoniazid, moxifloxacin linezolid, clofazamine amoxyclav, alarithromycin thioacetazone	– Assess for dehydration; initiate rehydration if indicated – Initiate anti-emetic therapy – Discontinue suspected agent if this can be done without compromising regimen – rarely necessary
Gastritis	PAS, moxifloxacin isoniazid, linezolid amoxiclav, clarithromycin thioacetazone	– H2-blockers, proton-pump inhibitors, or antacids – Stop suspected agent(s) for short periods of time (e.g., one to seven days) – Lower dose of suspected agent, if this can be done without compromising regimen – Discontinue suspected agent if this can be done without compromising regimen
Hepatitis	Isoniazid, PAS, moxifloxacin linezolid	– Stop all therapy pending resolution of hepatitis

		– Eliminate other potential causes of hepatitis – Consider suspending most likely agent permanently. Reintroduce remaining drugs, one at a time with the most hepatotoxic agents last, while monitoring liver function
Cutaneous and hypersensitivity reactions		
Hypersensitivity	PAS, linezolid, clarithromycin	– Withhold all drugs and treat symptomatically with antihistamines/ steroids till the reaction subsides. – Identify offending drug in severe forms – Attempt desensitization of the offending drug – Discontinue suspected agent and substitute with the reserve drug
Cutaneous	Linezolid, clofazamine	– Treat symptomatically with antihistamines till the reaction subsides. – Patient should be counseled on the skin discoloration with long term use of clofazamine
Psychiatric		
Depression	Socio-economic circumstances, chronic disease, moxifloxcin, isoniazid	– Improve socioeconomic conditions – Group or individual counseling – Initiate antidepressant therapy – Lower dose of suspected agent if this can be done without compromising the regimen – Discontinue suspected agent if this can be done without compromising regimen
Neurological		
Peripheral neuropathy	Isoniazid, linezolid	– Increase pyridoxine to maximum daily dose (200 mg per day) – Initiate therapy with tricyclic antidepressants such as amitriptyline. Non-steroidal anti-inflammatory drugs or acetaminophen may help alleviate symptoms – Lower dose of suspected agent, if this can be done without compromising regimen – Discontinue suspected agent if this can be done without compromising regimen
Convulsions	Isoniazid, moxifloxacin, linezolid	– Suspend suspected agent pending resolution of convulsions – Initiate anticonvulsant therapy (e.g. phenytoin, valproic acid) – Increase pyridoxine to maximum daily dose (200 mg per day). – Restart suspected agent or reinitiate suspected agent at lower dose, if essential to the regimen

		– Discontinue suspected agent if this can be done without compromising regimen
Hearing loss	Capreomycin clarithromycin	– Document hearing loss and compare with baseline audiometry if available – Change parenteral treatment to capreomycin if patient has documented susceptibility to capreomycin – Increase frequency and/or lower dose of suspected agent if this can be done without compromising the regimen(consider administration three times per week) – Discontinue suspected agent if this can be done without compromising the regimen
Ocular	Linezolid	In consultation with the ophthalmologist
Musculoskeletal		
Arthralgia	Moxifloxacin	– Initiate therapy with non-steroidal anti-inflammatory drugs. – Lower dose of suspected agent, if this can be done without compromising regimen. – Discontinue suspected agent if this can be done without compromising regimen
Musculoskeletal	Moxifloxacin	In consultation with the orthopedician
Haematological		
Haematalogic	Capreomycin, PAS, linezolid, clarithromycin	Monitor for anemia, bleeding tendency and cell counts
Renal		
Renal toxicity	Capreomycin linezolid clarithromycin	– Discontinue suspected agent – Consider using capreomycin if an aminoglycoside had been the prior injectable in regimen – Consider dosing 2 to 3 times a week if drug is essential to the regimen and patient can tolerate (close monitoring of creatinine) – Adjust all TB medications according to the creatinine clearance
Metabolic and Electrolytes		
Metabolic and electrolyte	Capreomycin, linezolid moxifloxacin	Periodic monitoring of serum electrolytes and blood sugar
Endocrine		
Thyroid dysfunction	PAS	Supplement with thyroxine
Cardiovascular		
QT prolongation	Moxifloxacin, clarithromycin	– Monitor for QT prolongation, monitor platelet count, bleeding tendencies

- **Chemoprophylaxis of contacts of MDR-TB cases**
  Among contacts of patients with MDR-TB, the use of isoniazid may reasonably be questioned. Although alternative prophylaxis treatments have been suggested, there is no consensus regarding the choice of the drug(s) and the duration of treatment. Prompt treatment of MDR-TB is the most effective way of preventing the spread of infection to others. The following measures should be taken to prevent spread of MDR-TB infection:
  - Early diagnosis and appropriate treatment of MDR-TB cases;
  - Screening of contacts as per RNTCP guidelines
  - Further research into effective and non-toxic chemoprophylaxis in areas of high MDR-TB prevalence.

Patient education

- Providing counseling and health education to the MDR-TB patient and their family members about the disease and about the necessity of taking regular and adequate treatment is of utmost importance.
- Health education and counseling is provided to all patients and family members at different levels of health care, right from one at the periphery to those at the DOTS-Plus site facility. It is started at the initial point of contact and carried on a continuous basis at all visits by the patient to a health facility. The counseling and motivation is required to be done not only of the patient but also of the family members.

EPILEPSY

A seizure is a paroxysmal event due to abnormal excessive or synchronous neuronal activity in the brain. Epilepsy describes a condition in which a person has recurrent seizures due to a chronic, underlying process.

Classification of Seizures

- Generalized seizures
  Tonic clonic, clonic, tonic, absence (typical/ atypical), atonic and myoclonic
  May be focal, generalized, or unclear epileptic spasms
- Focal seizures (can be further described as having motor, sensory, autonomic, cognitive, or other features) with and without dyscognitive features

Investigations

It should be made sure that the patient had a seizure and other differentials should be ruled out. In the patient with prior seizures or a known history of epilepsy, the evaluation is directed toward (1) underlying cause and precipitating factors (2) adequacy of current therapy.

Pharmacological treatment

- Treatment of underlying conditions
  - Metabolic imbalances such as abnormal serum electrolytes and glucose, medications (theophylline), illicit drug use (e.g., cocaine), structural CNS lesion such as a brain tumor, vascular malformation, or brain abscess.

- Avoidance of precipitating factors like sleep deprivation, alcohol, bright light, etc.
- Antiepileptic drug therapy mainstay of treatment for most patients with epilepsy.

Table 16. Selection of antiepileptic drugs according to classification of epilepsy

Generalized tonic clonic seizures	Focal seizures	Typical absence seizures	Atypical absence, myoclonic and atonic seizures
First line drugs			
Valproic acid	Lamotrigine	Valproic acid	Valproic acid
Lamotrigine	Capbamazepine	Ethosuximide	Lamotrigine
Topiramate	Oxcarbazepine		Topiramate
	Phenytoin		
	Levetiracetam		
Alternatives			
Zonisamide	Topiramate	Lamotrigine	Clonazepam
Phenytoin	Zonisamide	Clonazepam	Felbamate
Capbamazepine	Valproic acid		
Oxcarbazepine	Tiagabine		
Phenobarbitone	Gabapentin		
Primidone	Lacosamide		
Felbamate	Phenobarbitone		
	Primidone		
	Felbamate		

Pregnancy

- Most women with epilepsy who become pregnant will have an uncomplicated gestation and deliver a normal baby. Since the potential harm of uncontrolled convulsive seizures on the mother and fetus is considered greater than the teratogenic effects of antiepileptic drugs, it is currently recommended that pregnant women be maintained on effective drug therapy. Patients should take folic acid (1–4 mg/d), to prevent development of neural tube defects.

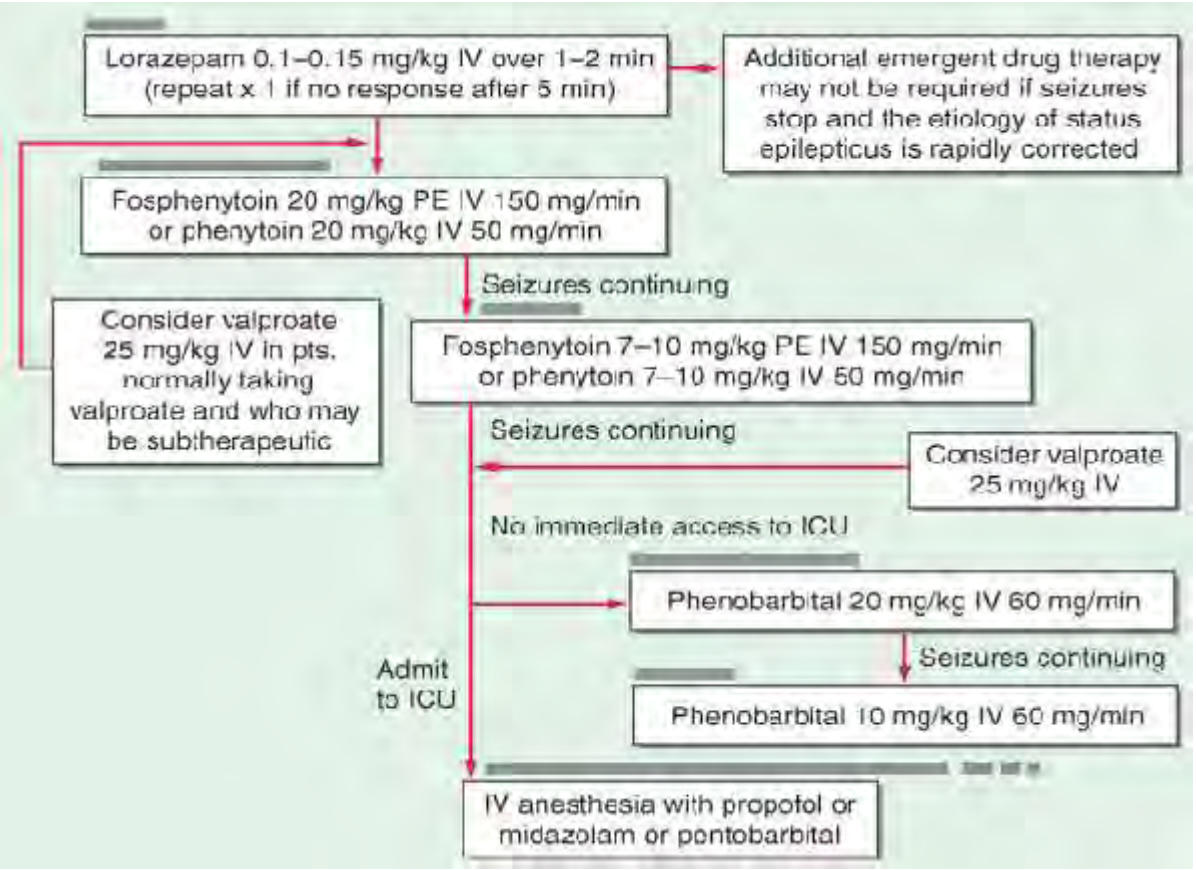
Contraception

- Drugs such as carbamazepine, phenytoin, phenobarbital, and topiramate can significantly decrease the efficacy of oral contraceptives via enzyme induction and other mechanisms. Patients should be advised to consider alternative forms of contraception.

STATUS EPILEPTICUS

Status epilepticus refers to continuous seizures or repetitive, discrete seizures with impaired consciousness in the interictal period. The duration of seizure activity sufficient to meet the definition of status epilepticus has traditionally been specified as 15–30 minutes. The treatment plan for an adult with status epilepticus is as in Figure 10.

Figure 10. Treatment algorithm for generalized tonic clonic status epilepticus in adults



URINARY TRACT INFECTION

Urinary tract infections include infection of any part of urinary tract from kidneys to urethra. It can be caused by unhygenic lifestyle, sexually transmitted, per urethral catheterization, renal calculus disease and vesicoureteric reflux.

Salient features

- Burning micturition, urinary frequency, flank pain, fever
- Predisposing factors are diabetes mellitus, large prostate, and circumcision.
- Common organisms are *E. coli, Pseudomonas, Klebsiella.*

Investigations

- TC with DC**,** X-ray KUB**,** USG-KUB**,** Urine (routine and microbiological examination)
- Urine culture and sensitivity to be done if bacterial colony count is more than 1000.

Non pharmacological treatment

Plenty of fluids orally, maintain hygiene

Pharmacological treatment

- Tab. ciprofloxacin 500 mg twice a day for five days.
- In resistance cases 3rd generation cephalosporins (cefixime) followed by antibiotics as per culture and sensitivity report.
- If fever then Tab. paracetamol 500 mg thrice a day for three days.

Chapter 3

EMERGENCY CONDITIONS

CARDIOPULMONARY RESUSCITATION

Cardiopulmonary resuscitation (CPR) consists of a series of maneuvers by which oxygenated blood supply to brain and vital organs is maintained during cardiopulmonary arrest (CPA) i.e. cessation of respiration and circulation. In children, CPA is not sudden but end result of long period of hypoxaemia secondary to inadequate ventilation, oxygenation or circulation. Therefore, prompt management of these is essential to prevent CPA, the outcome of which is poor.

Diagnosis of cardiopulmonary and cardiac arrest

1. Absence of pulse in major arteries (carotid or femoral in older children and femoral or brachial in infants as carotid is difficult to palpate due to short neck).
2. Absence of heart sounds on auscultation.
3. Asystole /ventricular fibrillation on ECG.

Respiratory arrest

Absence of respiration on examination (absent chest movements), listening (absent air flow on bringing ears in front of mouth) and feeling (absent air flow on keeping hands in front of mouth or nose).

Levels of CPR

Basic life support (BLS): The elements of CPR are provided without additional equipment. Skill and speed are most essential.

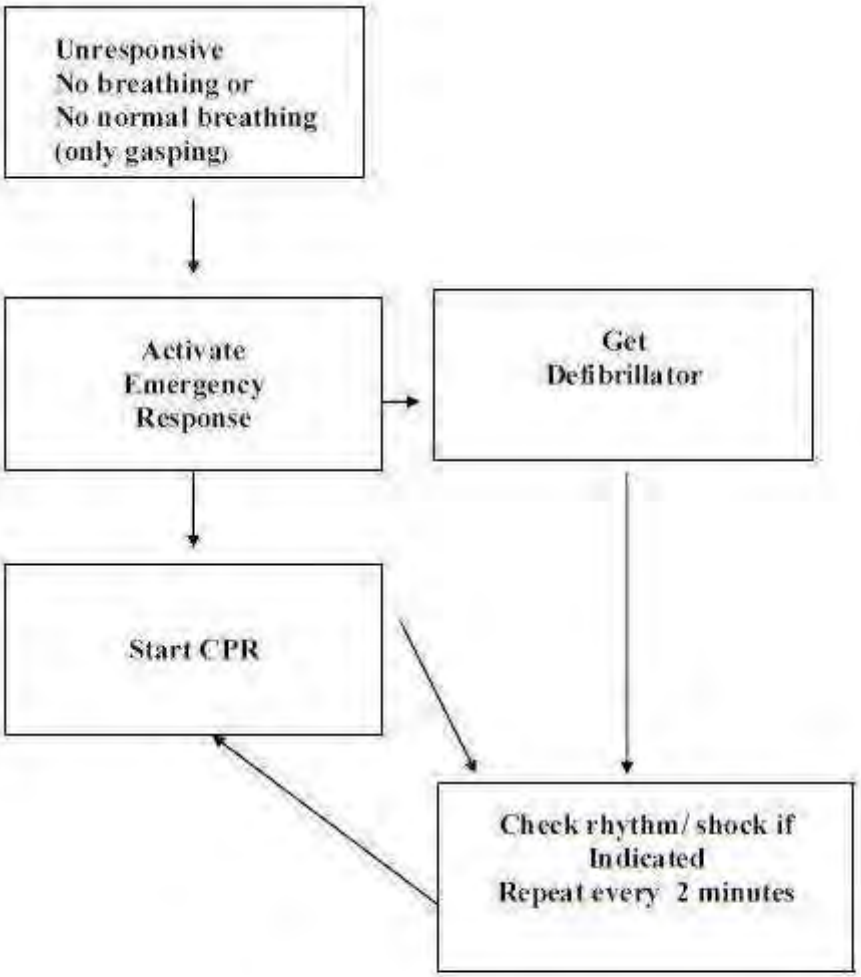
Advanced cardiac life support (ACLS): Use of equipment and drugs for assisting ventilation or circulation.

Primary survey to assess the patient's condition:

- Tap the patient on the shoulder and ask, "Are you all right?"
- If the patient does not respond, call for help.
- Give 2 minutes (5 cycles) of CPR before leaving patient to call for help.
- Check for pulse or any other signs of life; begin CPR, 30 compressions to 2 respirations (push hard and fast--100/minute and release chest completely).
- Open airway and check for breathing. Give 2 breaths that make the chest rise.
- Start an IV line.
- Begin oxygen.
- Attach a monitor
- Intravenous / Intra osseous (IO) access is a priority over advanced airway management. If an advanced airway is placed, change to continuous chest compressions without pauses for breaths. Give 8 to 10 breaths per minute and check rhythm every 2 minutes.

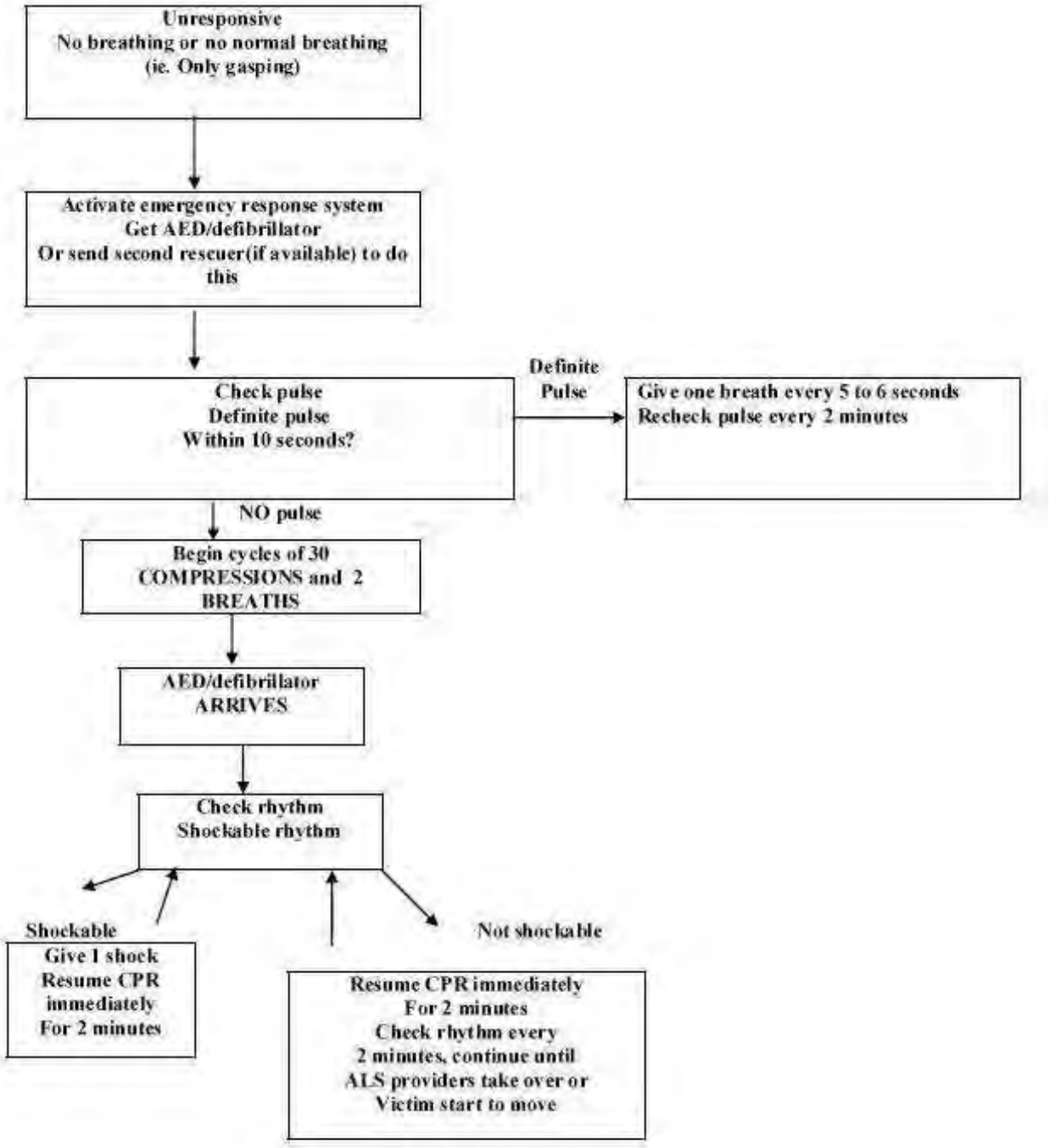
CHART

SIMPLIFIED ADULT BLS



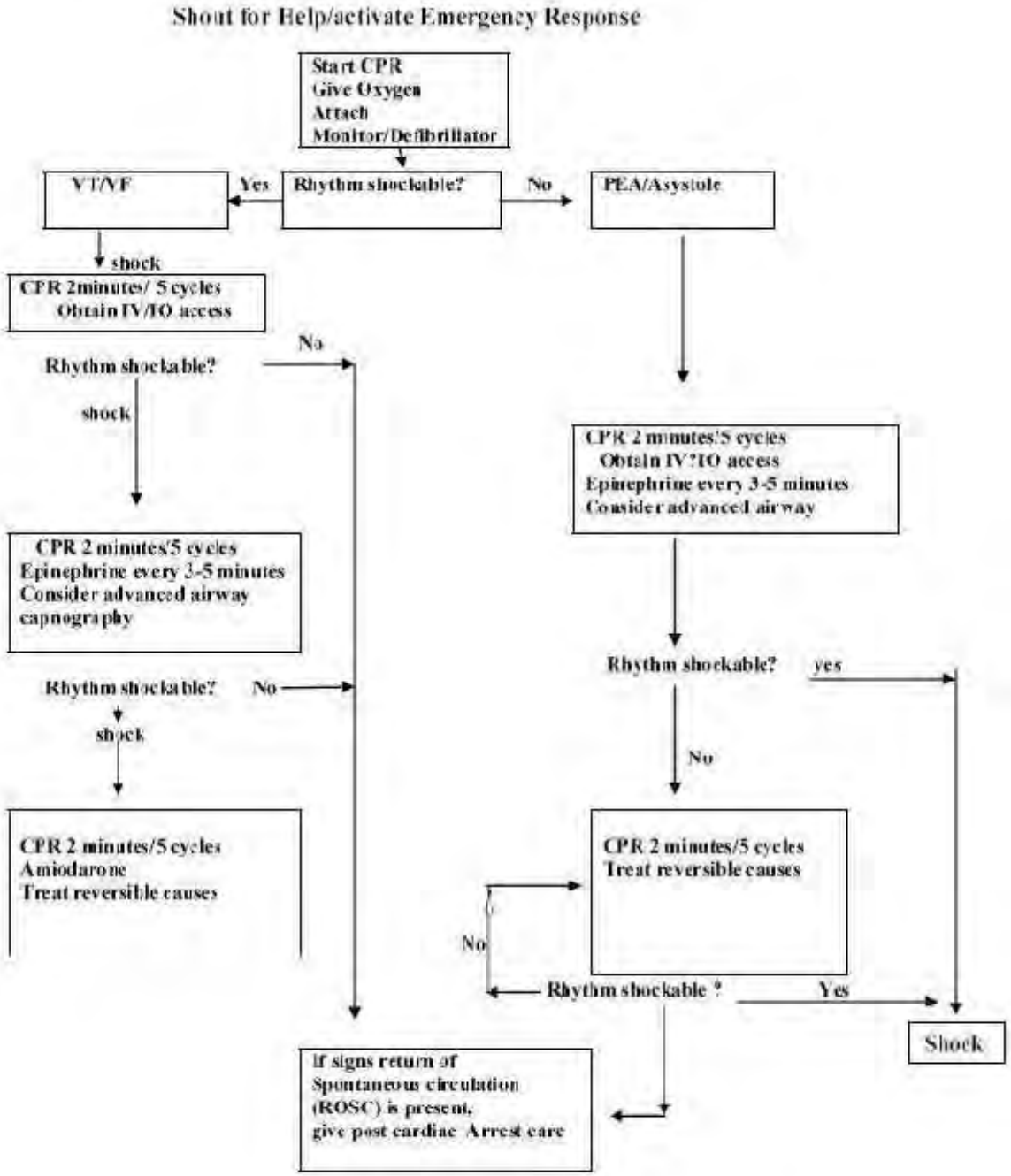
Source 2010 American Heart Association Guidelines

ADULT BLS HEALTHCARE PROVIDERS



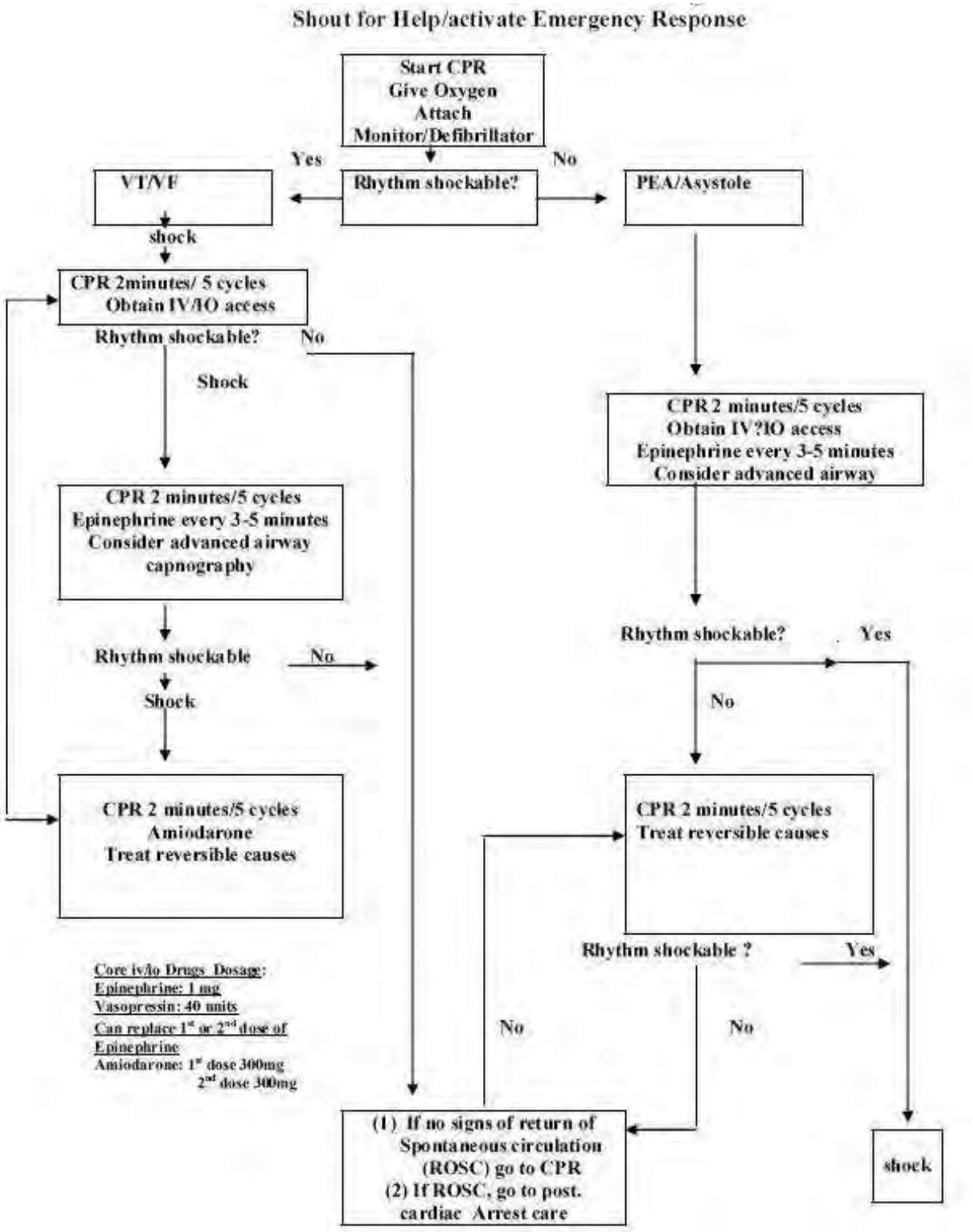
Source 2010 American Heart Association Guidelines

AHA ACLS ADULT CARDIAC ARREST ALGORITHM



Source 2010 American Heart Association Guidelines

AHA ACLS ADULT CARDIAC ARREST ALGORITHM



Source 2010 American Heart Association Guidelines

ANAPHYLAXIS

It is a generalized hypersensitivity reaction characterized by hypotension, peripheral circulatory collapse and respiratory difficulty in the form of stridor and dyspnoea. Anaphylaxis can be due to food, inhaled/ingested allergens or drugs. Symptoms occur instantaneously or within a few minutes following exposure to the offending agent. Anaphylaxis to oral drugs may take 1-2 hours, but in many patients it can be instantaneous. A severe anaphylactoid reaction is a life-threatening emergency. Effective treatment depends on prompt diagnosis and rapid supplementation of appropriate therapy.

Clinical criteria for diagnosis

Anaphylaxis is highly likely when any one of the following three criteria is fulfilled:

1. Acute onset of an illness (minutes to several hours) with involvement of the skin, mucosal tissue, or both (e.g., generalized urticaria, itching or flushing, swollen lips/tongue/uvula) with one of the following,
    - Respiratory compromise (e.g., dyspnoea, wheeze-bronchospasm, stridor, reduced peak expiratory flow (PEF), hypoxemia)
    - Reduced blood pressure or associated symptoms of end-organ dysfunction (e.g. hypotonia, collapse, syncope, incontinence)
    - Involvement of the skin-mucosal tissue
    - Persistent gastrointestinal symptoms (e.g., cramping abdominal pain, vomiting) or
2. Reduced blood pressure after exposure to known allergen,
    - Infants and children- low systolic blood pressure (age-specific) or greater than 30% decrease in systolic blood pressure
    - Adults- systolic blood pressure of less than 90 mm Hg or greater than 30% decrease from baseline

Salient features

Skin, subcutaneous tissue, and mucosa

- Flushing, itching, urticaria (hives), angioedema, morbilliform rash, piloerection
- Periorbital itching, erythema and edema, conjunctival erythema, tearing
- Itching of lips, tongue, palate, and external auditory canals; and swelling of lips, tongue, and uvula
- Itching of genitalia, palms, and soles

Respiratory

- Nasal itching, congestion, rhinorrhea, sneezing
- Throat itching and tightness, dysphonia, hoarseness, stridor, dry staccato cough
- Lower airways: increased respiratory rate, shortness of breath, chest tightness, deep cough, wheezing/bronchospasm, decreased peak expiratory flow
- Cyanosis, respiratory arrest

Gastrointestinal

- Abdominal pain, nausea, vomiting (stringy mucus), diarrhea, dysphagia

Cardiovascular system

- Chest pain
- Tachycardia, bradycardia (less common), other arrhythmias, palpitations
- Hypotension, feeling faint, urinary or fecal incontinence, shock
- Cardiac arrest

Central nervous system

- Aura of impending doom, uneasiness (in infants and children, sudden behavioral change, e.g. irritability, cessation of play, clinging to parent); throbbing
- Headache (pre-epinephrine), altered mental status, dizziness, confusion, tunnel vision

Non pharmacological treatment

- Remove exposure to trigger if possible (discontinue intravenous diagnostic or therapeutic agent that seems to triggering symptoms). Place the patient supine and elevate the extremities.
- Observe vital signs frequently and if possible, monitor electrocardiogram and pulse oximetry.
- For severe anaphylaxis with shock, initial management should be directed at the ABC of resuscitation, namely maintenance of adequate airway - suction, breathing, and circulation. If working alone, call for assistance.

Pharmacological treatment

- Inj. adrenaline 1:1000 (1mg/ml), 0.01 ml/kg (maximum 0.3 ml in children and 0.5 ml in adults) IM (mid anterolateral aspect of thigh). If necessary, dose can be repeated every 15 minutes.

- If the anaphylaxis is due to an allergen extract or to a hymenoptera sting into an extremity, half the dose of adrenaline can be infiltrated locally, subcutaneously after dilution with 2 ml saline.
- High flow oxygen (4-6 lit/min by face mask or oropharyngeal airway).
- Establish two wide bore intravenous lines. Commence rapid fluid resuscitation with Inj. NS (0.9% normal saline) at rate of 5-10 ml/kg in 5 to 10 minutes. One to two litres of fluid bolus can be given.
- Intubate and commence intermittent positive pressure ventilation if severe laryngeal obstruction, bronchospasm, circulatory shock or coma.
- If no response to the initial intramuscular dose of adrenaline, administer adrenaline 5 mcg/kg slowly in to the intravenous line. Repeat at 5 minutes intervals depending on response. If the patient remains in shock, start an adrenaline infusion (preferably via a central venous line), commencing at 0.1 mcg/kg/min in children and 0.2 mcg/kg/min in adults with titration required to restore blood pressure. Large doses of adrenaline may be needed.
- Commence cardiopulmonary resuscitation, if necessary.
- If the only manifestation of anaphylaxis is urticaria or angioedema, initial IM dose of adrenaline should be given in addition to antagonists. If no other manifestation, patient can be kept under observation for at least 12 hours and then discharged.

Additional measures

- IV infusion of chlorpheniramine 10 mg (adult), 2.5-5 mg (child) or diphenhydramine 25-50 mg (adult) (1 mg/kg, maximum 50 mg [child]).
- Inj. salbutamol solution, 2.5 to 5 mg/3 ml given by nebulizer and face mask.
- IV hydrocortisone infusion 200 mg (adult), maximum 100 mg (child); or methylprednisolone 50-100 mg (adult); 1 mg/kg, maximum 50 mg (child).

Supportive treatment

All patients with severe anaphylactic reaction must be hospitalized. Patients who remain clinically unstable after initial resuscitation should be admitted to intensive care unit (ICU). Patients responding to the initial treatment should be observed for possible late reaction.

Patient education

To check and look for the cause (food, drugs etc.) and to avoid it in future

ACUTE AIRWAY OBSTRUCTION

The cause of acute airway obstruction can be underlying malignancy, abscess, laryngeal cyst or laryngocele. It is necessary to assess the severity of the condition before initiating the treatment.

Emergency treatment

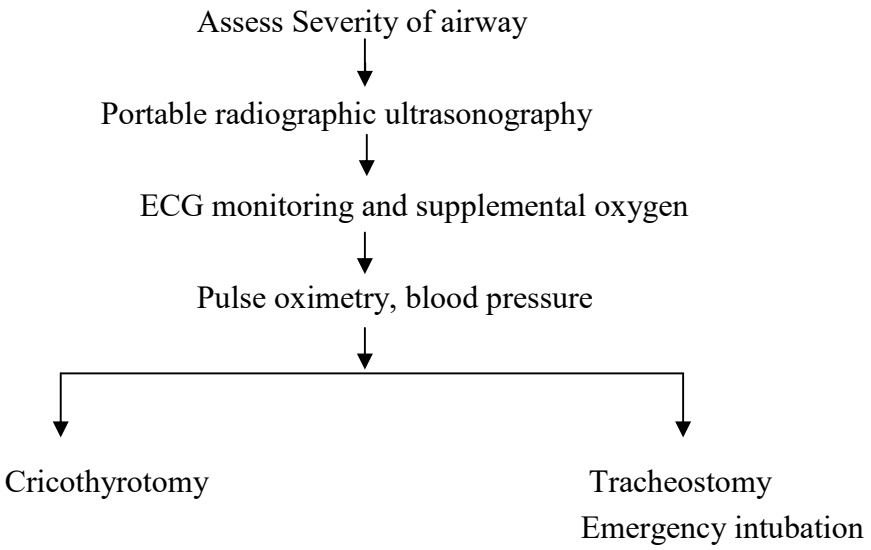
- Head extension with lower jaw thrust forward
- Suctioning for clearing secretions

- Ventilatory mask with 100% oxygen
- Endotracheal intubation or tracheostomy

In stable patient

- Plain X-ray neck
- Fibre optic endoscopy / micro-laryngoscopy / bronchoscopy to find out the cause

Treatment guidelines



STRIDOR

Stridor is noisy, harsh and difficult breathing due to obstruction of the airway. It is classified inspiratory, expiratory and biphasic type. Inspiratory stridor occurs due to an obstruction at or above the level of vocal cords. Expiratory stridor is due to obstruction at bronchiolar level such as bronchial asthma while biphasic stridor occurs in tracheal and subglottic stenosis. In children the stridor of laryngo-tracheo-bronchitis has a crowing quality and is called 'Croup'.

Salient features

- Restlessness, anxiety and distress, labored / noisy breathing, predominantly on inspiration
- Difficulty in swallowing and speaking
- In children: cyanosis, chest retraction, tachypnea, altered level of consciousness, feeding difficulties

Pharmacological treatment

- Humidified oxygen
- Inj. hydrocortisone 100 mg IV stat and can be repeated as required.
- Inj. ampicillin 500mg QID IV/for 5-7 days
- Inj. adrenaline 0.2-0.5mg subcutaneously if necessary.

SHOCK

Shock is defined as a circulatory insufficiency that creates an imbalance between tissue oxygen supply and demand.

Classification and causes of shock

1. **Haemorrhagic shock**
   - Traumatic – blunt or penetrating injury, fractures of long bones and pelvis
   - Non-traumatic - GI bleeds (e.g. peptic ulcer, gastric mucosal erosions, oesophageal varices, typhoid bleeds, disseminated intravascular coagulation (DIC), aortic dissection, rupture of aneurysm of a large vessel
2. **Hypovolaemic shock**
   - Fluid loss from vomiting and/or diarrhoea e.g. in cholera, other GI infections
   - Fluid loss in diabetes mellitus, adrenal insufficiency, excessive sweating, exfoliative dermatitis, diabetes insipidus, reaccumulation of ascites after tapping
   - Sequestration of fluid e.g. in intestinal obstruction, pancreatitis, burns, crush injuries
3. **Cardiogenic shock**
   - Acute myocardial infarction
   - Cardiomyopathy, cardiac arrhythmias
   - Mechanical causes e.g. valvular disease, outflow tract obstruction, ruptured ventricular septum
4. **Distributive or vasogenic (relative hypovolaemia)**
   - Septic shock; toxic shock syndrome
   - Anaphylactic
   - Neurogenic

Shock is a progressive disorder, if untreated can lead to severe haemodynamic and metabolic deterioration causing multi-organ failure. Stages of shock can be arbitrarily classified as:

- **Early compensated shock** - Vital organ function is maintained by intrinsic compensatory mechanisms. Blood pressure is usually normal, there is increasing tachycardia and hypotension. The skin is cold and clammy, increased capillary refill time (>3 sec). If there is delay in treatment it may-lead to decompensated shock.
- **Decompensated shock** - There is fall in blood pressure and cardiac output. Features of peripheral poor perfusion are compounded with manifestations of vital organ impairment. Patient may have alteration of mentation (impaired cerebral perfusion), oliguria (renal hypoperfusion) and myocardial ischaemia (coronary flow impairment). Patient has acrocyanosis, cold and damp extremities and a pale look. If untreated, progress to irreversible state of shock.
- **Irreversible shock** is a term applied to the clinical situation in which even haemodynamic correction does not halt the progressive organ failure.

Salient features

- Tachycardia, blood pressure may be normal or elevated due to compensatory mechanisms, later falls as cardiovascular compensation fails. Neck veins may be distended or flattened, depending on the etiology of shock. Decreased coronary perfusion pressures can lead to ischemia, decreased ventricular compliance, and increased left ventricular diastolic pressure and pulmonary edema.
- Tachypnea, increased minute ventilation, and increased dead space are common. Bronchospasm, hypocapnia with progression to respiratory failure, and adult respiratory distress syndrome can be seen.
- Skin pale, dusky, clammy skin with cyanosis, sweating, altered temperature, and decreased capillary refill, oliguria.

Treatment

- Start oxygen 4-6 L/min
- Airway control, endotracheal intubation
- Adequate venous access. Large-bore peripheral intravenous catheters for adequate fluid resuscitation. Central venous access may be necessary for monitoring and employing pulmonary artery catheters, venous pacemakers, and long-term vasopressor therapy.
- Volume replacement. Isotonic, intravenous crystalloid fluids (0.9% NaCl, Ringer's lactate) are preferred for the initial resuscitation phase. Initial bolus volume is 20 to 40 ml/kg over 10 to 20 min (rapid infusion). Colloids (dextran, hetastarch, etc.) can be used till blood is available.
- Blood is the ideal resuscitative fluid for hemorrhagic shock or in the presence of significant anemia. Fully cross-matched blood is preferred, but if more rapid intervention is required, type specific or type O negative blood may be employed. The decision to use platelets or fresh frozen plasma (FFP) should be based on evidence of impaired hemostasis and on frequent monitoring of coagulation parameters. Platelets are generally given if there is ongoing hemorrhage and the platelet count is 50,000 or less; FFP is indicated if the prothrombin time is prolonged more than 1.5 times.
- Vasopressors should be used if there is persistent hypotension after adequate volume resuscitation. Inj. dobutamine 2.0 to 20.0 µg/kg/min for systolic BP over 100 mmHg, dopamine 2.5 to 20.0 µg/kg/min for systolic BP 70 to 100 mmHg, and Inj. norepinephrine 0.5 to 30.0 µg/min for systolic BP under 70 mmHg.
- Acidosis should be treated with adequate ventilation and fluid resuscitation. Use of sodium bicarbonate (1 meq/kg) is controversial. If it is used, it is given only in the setting of severe acidosis refractory to ventilation and fluid resuscitation.
- Early surgical or medical consultation for admission or transfer as indicated.

Monitoring

Special care in ICU should be taken by careful and continuous monitoring of BP, pulse rate and respiratory rate, urine output and mental status frequently. Sedated patients should be allowed to awaken ("drug holiday") daily to assess their neurologic status and to shorten duration of ventilator support.

Management guidelines for common causes of shock

Hemorrhagic Shock

- Ensure adequate ventilation/oxygenation
- Provide immediate control of hemorrhage (e.g., traction for long bone fractures, direct pressure)
- Initiate judicious infusion of isotonic crystalloid solution (10–20 ml/kg) or colloid solution
- With evidence of poor organ perfusion and 30-minute anticipated delay to hemorrhage control, begin packed red blood cell (PRBC) infusion (5–10 ml/kg)
- Treat coincident dysrhythmias (e.g., atrial fibrillation with synchronized cardioversion)

Cardiogenic Shock

- Ameliorate increased work of breathing; provide oxygen and positive end-expiratory pressure (PEEP) for pulmonary edema
- Begin vasopressor or inotropic support; Inj. norepinephine (0.5 µg/min) and Inj. dobutamine (5 µg/kg/min)
- Seek to reverse the insult (e.g., initiate thrombolysis, arrange percutaneous transluminal angioplasty)
- Consider intra-aortic balloon pump counterpulsation for refractory shock

Septic Shock

- Ensure adequate oxygenation
- Administer 20 ml/kg of crystalloid (0.9% normal saline or RL) or 5 ml/kg of colloid (hemaccel or hetastarch), and titrate infusion to adequate central venous pressure and urine output
- Begin antimicrobial therapy; attempt surgical drainage or débridement
- Begin packed RBC infusion for hemoglobin < 8 g/dL
- If volume restoration fails to improve organ perfusion, begin vasopressor support; initial choice includes dopamine infusion at 5–15 µg/kg/min

FLUID AND ELECTROLYTE IMBALANCE AND REPLACEMENT (IN ADULTS)

Disturbances in fluid and electrolyte balance occur in a wide spectrum of diseases such as following burns, trauma and major surgery. The conventional and easy method of evaluating disturbances in fluid and electrolyte balance is the frequent measurement of the concentration of serum electrolytes. It is crucial to remember that intracellular and extracellular electrolytes are normally constant, and that major shifts in and out of 'compartments' can occur in disease with minimal changes in serum electrolyte. Compositional changes also involve disturbances in acid-base balance.

I. Volume deficit

- **Obvious causes**

Vomiting, diarrhoea, intestinal fistulae, nasogastric suction, fluid loss following burns, sequestration of fluid in soft tissue injuries and infections, diuretics, renal disease/adrenal insufficiency

- **Less obvious causes**

Inadequate fluid intake, fluid loss through excessive sweating as in high fever, hot humid temperature, haemodialysis, haemofiltration from surgical incisions and tetanus

Treatment

The first principle is to restore circulating volume through infusion of intravenous fluids. Once this is satisfactorily achieved, disturbances in electrolytes and acid-base balance if present need to be rectified. Various fluids used for volume replacement are given below:

Replacement fluids

- Replacement fluids are used to replace abnormal loss of blood, plasma or other extracellular fluids as first line treatment for hypovolemia in:
  - Patients with established hypovolemia e.g. haemorrhagic shock.
  - Maintenance of normovolaemia in patients with ongoing fluid losses e.g. surgical blood loss.
- Intravenous replacement fluids are the first line of treatment for hypovolemia.
- Crystalloid maintenance fluids, which contain dextrose, are not suitable for use as replacement fluids. However, crystalloid solutions with a similar concentration of sodium to plasma (normal saline or balanced salt) solutions (Ringer's lactate or Hartmann's solutions) are effective as replacement fluids. These should be available in all hospitals where intravenous replacement fluids are used. Crystalloids should be infused in a volume at least three times the volume lost in order to correct hypovolemia.
- All colloid solutions (albumins, dextran, gelatins and hydroxyethyl starch solutions) are replacement fluids. However, they have not been shown to be superior to crystalloids in resuscitation. Colloid solutions should be infused in a volume equal to the blood volume deficit.
- Plasma should never be used as a replacement fluid.
- Plain water should never be infused intravenously as it can cause haemolysis.

- In addition to the intravenous route, the intraosseous, oral, rectal or subcutaneous routes can be used for the administration of fluids, blood and certain drugs. However, with the exception of intraosseous route, other routes are generally unsuitable in severely hypovolemic patients.
- Rectal fluids are administered through a plastic or rubber enema tube which is inserted into the rectum and connected to a bag or bottle of fluid. A safe and effective solution for a rectal rehydration is 1 liter of clean drinking water with teaspoon of table salt.
- Subcutaneous fluids can be used when other route of administration of fluids are unavailable. A cannula or needle is inserted into the subcutaneous tissue (the abdominal wall is a preferred site) and sterile fluids are administered in a conventional manner. Do not give dextrose-containing solutions subcutaneously as they can cause sloughing of tissues.
- Oral and nasogastric fluids: Oral rehydration can often be used in mildly hypovolemic patients if the oral route is not contraindicated. Do not use if:
  - The patient is unconscious.
  - The patient has gastrointestinal lesions or reduced gut motility e.g. obstruction.
  - General anaesthesia and surgery is planned imminently.

WHO/UNICEF formula for oral rehydration fluid:

Dissolve in one litre of drinkable water

Sodium chloride	2.6 g/L
Trisodium citrate, dihydrate	2.9 g/L
Potassium chloride	1.5 g/L
Glucose anhydrous	13.5 g/L

Resulting concentrations

Na 75 mmol/L, CI 65 mmol/L, K 20 mmol/L, Glucose anhydrous 75 mmol/ L, Citrate 10 mmol/L, total osmolarity 245 mmol/L.

Maintenance fluids

Maintenance fluids are used to replace the normal physiological loss that occurs through skin, lung, faeces and urine. Since a considerable proportion of the loss is water, maintenance fluids are mainly composed of water in the form of a dextrose solution. Some electrolytes may also be included in these solutions.

Before giving any intravenous infusion:

- Check that the seal of the infusion fluid bottle or bag is not broken.
- Check the expiry date.
- Check that the solution is clear and free from visible particles.

II. Volume excess

Volume excess is often iatrogenic when the fluid intake has consistently exceeded the output. Excessive intravenous infusions of saline and blood transfusions are important

causes of hypervolemia. Renal insufficiency, congestive heart failure, liver disease and other causes of sodium retention, or excessive sodium administration can all produce increase in extracellular fluid content and hypervolemia.

Salient features

- Oedema, ascites, pleural effusion, neck veins full, pulmonary congestion, hyperdynamic circulation with tachycardia, a warm skin, and a bounding pulse.
- Increase in the systolic pressure and pulse pressure.
- Central venous Pressure (CVP) > 12 mmHg, Pulmonary Capillary Wedge Pressure (PCWP) >20 mmHg in the presence of pulmonary oedema.

Treatment

In moderate volume excess, salt restriction, restriction of fluid intake and the use of frusemide as a diuretic will solve the problem. Fulminant pulmonary oedema secondary to over hydration from over transfusion of blood or fluids is more appropriately dealt with by phlebotomy in stages so that pulmonary capillary wedge pressure is reduced below 15 mm Hg. Rarely ultrafiltration (dialysis) may be required.

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2. Drugs Used in Anaesthesia. In: WHO Model Prescribing Information, 1989, World Health Organization.
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SEPTICEMIA

Septicemia is a clinical syndrome caused by invasion of bloodstream by microorganisms. Gram-negative and gram-positive bacteria account for most common sepsis cases.

Definitions

- Bacteremia (fungemia): presence of viable bacteria (fungi) in the blood, as evidenced by positive blood cultures.
- Systemic Inflammatory Response Syndrome (SIRS*)*: at least two of the following conditions: (1) oral temperature of >38° C or <35° C, (2) respiratory rate of >20 breaths/min or partial pressure of arterial carbon dioxide ($PaCO_2$) of <32 mm Hg, (3) heart rate of >90 beats/min, (4) leukocyte count of >12,000/dl or <4000/dl or >10% bands.
- Severe sepsis: sepsis with one or more signs of organ dysfunction, hypoperfusion, or hypotension, such as metabolic acidosis, alteration in mental status, oliguria, or adult respiratory distress syndrome.

- Septic shock: sepsis with hypotension that is unresponsive to fluid resuscitation plus organ dysfunction or perfusion abnormalities as listed for severe sepsis.
- Multiple Organ Dysfunction Syndrome (MODS*)*: dysfunction of more than one organ, requiring intervention homeostasis.

Salient features

- Fever or hypothermia may be seen in sepsis. Hypothermia is more often seen in patients at the extremes of age and in immunocompromised patients.
- Other signs include tachycardia, wide pulse pressure, tachypnea, and hypotension.
- Mental status changes ranging from mild disorientation to coma are commonly seen.
- Ophthalmic manifestations include retinal hemorrhages, cotton-wool spots, and conjunctival petechiae.
- Cardiovascular manifestations initially include vasodilation, resulting in warm extremities.
- Cardiac output is maintained early in sepsis through a compensatory tachycardia. As sepsis progresses, hypotension may occur. Patients in septic shock may demonstrate a diminished response to volume replacement.
- Renal manifestations include azotemia, oliguria, and active urinary sediment due to acute tubular necrosis.
- Hepatic dysfunction present as cholestatic jaundice, increases in transaminases, alkaline phosphatase, and S.bilirubin. Painless mucosal erosions may occur in the stomach and duodenum and cause upper GI bleeds.
- Skin findings include cellulitis, erysipelas, and fasciitis. Hypotension and disseminated intravascular coagulation (DIC) can also produce skin changes, including acrocyanosis and necrosis of peripheral tissues.
- Hematologic changes include neutropenia, neutrophilia, thrombocytopenia, and DIC. Thrombocytopenia occurs in over 30 percent of patients with sepsis. DIC is more often associated with gram-negative sepsis. Decompensated DIC presents with clinical bleeding and thrombosis. Laboratory studies can show thrombocytopenia, prolonged prothrombin time (PT) and partial prothromboplastin time (PTT), decreased fibrinogen level and antithrombin levels, and increased fibrin monomer, fibrin split values, and D-dimer values.
- Hyperglycemia may be the result of increased catecholamines, cortisol, and glucagon. Increased insulin resistance, decreased insulin production, and impaired utilization of insulin may further contribute to hyperglycemia.
- Arterial blood gas (ABG) studies in early sepsis may reveal hypoxemia and respiratory alkalosis. As perfusion worsens and glycolysis increases, a metabolic acidosis results.

Non pharmacological treatment

- Aggressive airway management with high-flow oxygen and endotracheal intubation
- Rapid infusion of crystalloid IV fluid (Ringer's lactate or normal saline) at 500 ml (20 ml/kg in children) every 5 to 10 min; 4 to 6 L (60 ml/kg in children)

- Remove source of infection-remove indwelling IV catheters and send the tips for quantitative culture. Replace urinary drainage catheters if obstructed. Consult the appropriate specialist for drainage of intra-abdominal, sinus, or soft tissue abscesses.
- Monitor blood pressure, pulse, capillary refill, central venous pressure, pulmonary capillary wedge pressure, and urine output (>30 ml/h in adults, 1 ml/kg/h in children) to evaluate therapy. If blood loss is suspected, blood replacement may be necessary

Pharmacological treatment

- Inj. dopamine 5 to 20 µg/kg/min, titrated to response, if hypotension is refractory to IV fluid.
- If blood pressure remains <70 mmHg despite preceding measures, Inj. norepinephrine 8-12 µg/min loading dose and a 2-4 µg/min infusion to maintain mean arterial blood pressure of at least 60 mmHg should be started.
- Empiric antibiotic therapy. Dosages should be the maximum allowed and given intravenously. When source is unknown, therapy should be effective against both gram-positive and gram-negative organisms.
- In adults, a third-generation cephalosporin (ceftriaxone 1 g IV, cefotaxime 2 g IV, or ceftazidime 2 g IV) or an antipseudomonal beta lactamase–susceptible penicillin can also be used. Addition of an aminoglycoside (gentamicin 2 mg/ kg IV, tobramycin 2 mg/kg IV) to this regimen is recommended.
- In immunocompromised adults, ceftazidime 2 g IV, imipenem 750 mg IV, or meropenem 1 g IV alone is acceptable. If gram-positive infection is suspected (indwelling catheter or IV drug use), oxacillin 2 g IV or vancomycin 15 mg/ kg IV should be added. If an anaerobic source is suspected (intraabdominal, genital tract, odontogenic, and necrotizing soft tissue infection), metronidazole 7.5 mg/kg IV or clindamycin 900 mg IV should additionally be administered. If *Legionella* is a potential source, erythromycin 500 mg IV should be added.
- Acidosis to be treated with oxygen, ventilation, and IV fluid replacement. If acidosis is severe, administration of sodium bicarbonate 1 meq/kg IV is acceptable as directed by ABGs.
- DIC should be treated with fresh-frozen plasma 15 to 20 ml/kg initially to keep PT at 1.5 to 2 times normal and treated with platelet infusion to maintain serum concentration of 50 to 100,000.
- Inj. hydrocortisone (100 mg IV) for adrenal insufficiency

ORGANOPHOSPHOROUS POISONING

Organophosphorous (OP) insecticides are highly lipid soluble and are rapidly absorbed via dermal, gastrointestinal and respiratory routes. Common agents for OP poisoning are malathion, parathion, chlorpyrifos, dichlorvos, methylparathion which are used as insecticides.

Salient features

- Clinical features include dyspnoea, tightness around chest, nausea, vomiting, diarrhea, abdominal pain, excessive sweating and salivation, miosis, blurred vision, lacrimation, headache, giddiness, drowsiness, bradycardia, arrhythmias, QT prolongation, hypotension, fasciculation, muscle weakness, seizures and coma.
- Diagnosis can be confirmed by red blood cell (RBC) cholinesterase and plasma cholinesterase level which is reduced to 20% of the normal values.

Non pharmacological treatment

- Maintain patent airway: oral airway/intubation/mechanical ventilation if required
- Monitor vitals- temperature, pulse, blood pressure, respiratory rate
- Start intravenous line placement: maintenance IV fluids
- Decontamination: remove of all clothes
- Gastric lavage with activated charcoal and sample collection
- Wash skin with soap and water

Pharmacological treatment

- Inj. atropine 2 mg IV bolus, repeat every 5-10 minutes until atropinization (pulse >80/min, pupillary dilation and drying of secretions) followed by atropine IV infusion at 0.02-0.08 mg/kg/min to maintain atropinasation (glycopyrronium can be used in place atropine to avoid CNS toxicity).
- Inj. pralidoxime IV 30mg/kg immediately followed by 8 mg/kg/hr continued for at least 24 hours. If an infusion cannot be used, 30 mg/kg every 4 hourly.
- Inj. obidoxime 4 mg/kg bolus initially followed by 0.5 mg/kg/hr infusion Alternatively, 4 mg/kg bolus followed by 2 mg/kg every 4 hourly.
- IV crystalloids: normal saline/ringer's lactate.
- Benzodiazepines: midazolam/lorazepam/diazepam for treatment and prophylaxis of seizures

KEROSENE AND PETROL POISONING

Salient features

- Most common accidental poisoning in children.
- Inhalational of vapours or pulmonary aspiration can cause toxicity. The lethal dose is 30-100 ml.
- Mild- breathing difficulty, throat pain, nausea / vomiting, vision loss, low blood pressure, convulsions, headache, weakness, burns / irritation.
- Severe: pulmonary edema, hemoptysis, mental confusion, hallucination, stupor, cyanosis, convulsions, coma, death (due to ventricular fibrillation / respiratory failure).

Investigations

Chest x ray: perihilar densities, basal pneumonitis, atelectasis, consolidation

Treatment

- Remove all contaminated clothes of the patient
- Flush the skin and oral cavity for at least 15 minutes
- If the chemical is swallowed, immediately give the person water or milk
- If the person breathed in the poison, immediately move him or her to fresh air with patent airway with head low position
- Do not induce emesis
- Put the patient on monitor for vital signs – temperature, pulse, blood pressure, respiratory rate
- Provide oxygen through nasal cannula or face mask
- Take IV line – start intravenous crystalloids as a supportive therapy
- Patients should be monitored for respiratory complications.

DATURA POISONING

Datura stramonium (thorn apple) grows in India at high altitudes. The seeds and fruits are the most poisonous parts of the plant with hyoscine, hyoscyamine and traces of atropine, as the active principles.

Salient features

- Gastric irritation within half an hour of taking the poison.
- Bitter taste, dry mouth and throat, burning pain in the stomach and difficulty in swallowing and talking.
- Giddiness, ataxia, incoordination of muscles, flushed appearance of the face, dry hot skin, rise in temperature, diplopia, dilated pupils with loss of accommodation, reddening of the conjunctiva and drowsiness. Sometimes, an erythematous rash appears all over the body.
- Full, bounding pulse which later becomes weak and irregular.
- Muttering delirium, tries to run away from the bed, picks at bed clothes, tries to pull imaginary threads from the tips of his fingers and develops dreadful hallucinations of sight and hearing. The condition may pass on to stupor, convulsions, coma and sometimes death from respiratory failure.
- Death may occur within 4-24 hours.

Non pharmacological treatment

- Stomach wash with 1:10,000 potassium permanganate solution or 5% tannic acid solution.

- Decontamination of the patient must be individualized, based on time since exposure, type of anticholinergic agent involved, route of exposure, probable amount ingested, and severity of the clinical condition.
- Early gastric emptying (within 1 hour) may be considered for large ingestions, when an asymptomatic patient presents immediately after ingestion.
- Measure core body temperature with a flexible rectal probe. Aggressive temperature reduction with ice water or evaporative cooling with mist and fans should be the first priority in the severely hyperthermic patient. Antipyretic agents and simple cooling blankets are ineffective.

Pharmacological treatment

- Benzodiazepines like midazolam or lorazepam can be used to control agitation and sedation.
  Inj. midazolam 0.01-0.1 mg/kg IV or Inj. lorazepam 0.05-0.1 mg/kg IV
- In severe poisoning, Inj. physostigmine 1.2 mg IM or IV, to be repeated after half an hour, if necessary. Watch for side effects (bradycardia, heart block, excessive secretions).
- Inj. diazepam 10 mg IV diluted slowly may be given for convulsions.

ABDOMINAL INJURY

Abdominal injuries are quite common in this new era and it is must for a surgeon to quickly assess and promptly manage such injuries which can be fatal. They can be broadly classified as penetrating abdominal injury and blunt abdominal injury.

I. Penetrating abdominal injury

- Penetrating abdominal injuries caused by sharp weapon injury, missile injury/gunshot injury etc
- History of time and place of injury, nature of injury, direction of injuring weapon
- Physical examination assess vitals, assess wound thoroughly, look for peritoneal breach

Investigations

- Generally no investigations are required before proceeding for laparotomy.
- X-ray abdomen standing can be done in cases where peritoneal breach is in doubt

Treatment

- Resuscitate the patient, secure airway, maintain breathing, maintain circulation, arrange for compatible blood.
- Emergency exploratory laparotomy and further management according to organ injured.

II. Blunt abdominal injury

Blunt abdominal injuries can be due to assault by blunt objects, road traffic accidents and falls.

Investigations

- Haemoglobin levels, hematocrit, X-ray abdomen standing

- USG- abdomen, CECT- abdomen and pelvis.

Treatment

- Resuscitate the patient, secure airway, maintain breathing, maintain circulation, arrange for compatible blood.
- Temperature, pulse, respiratory rate, blood pressure (TPRBP) charting, abdominal girth (AG) charting.
- Management of blunt abdominal trauma depends upon vitals of the patients and organs involved as per seen in USG- Abdomen and Contrast Enhanced Computed Tomography (CECT) abdomen.
- All patients whose vitals are unstable are to be explored. Emergency exploratory laparotomy and followed by management according to organ involved.
- All patients whose vitals are stable are reviewed by CECT-abdomen and pelvis and managed according to organs involved and grade of injury.

FOREIGN BODY IN EAR

Occasionally a fly or an insect gets trapped in the external canal and cause distress and discomfort. The foreign body could be animate-insects or inanimate, organic -seeds or inorganic -stones, beads, cotton buds, match sticks etc.

Treatment

- Insects - immobilized using oil and remove with forceps
- Vegetable foreign bodies – remove with foreign body hook (syringing is not advisable).
- Non-organic foreign bodies- remove with syringing and foreign body hook.
- Impacted foreign bodies- remove under general anesthesia using the microscope

CHEMICAL INJURIES OF EYE

Salient features

- The common agents causing chemical injuries are alkalis (lime, cement, plasters), acids, solvents, detergents and irritants. Alkalis tend to penetrate more deeply than acids, as the latter coagulate surface proteins, forming a protective barrier.
- Symptoms include redness, pain, photophobia, blepharospasm and watering of eyes.
- Corneal epithelial defects ranging from scattered superficial punctate keratopathy (SPK), to focal epithelial loss, to sloughing of the entire epithelium may occur.
- Focal areas of conjunctival chemosis, hyperemia, hemorrhages, or a combination of these
- Mild eyelid edema, mild anterior chamber (AC) reaction and first- and second-degree burns of the periocular skin.

Investigations

- History: Time of injury? Specific type of chemical? Time between exposure and irrigation of eyes? Duration and type of irrigation? Was the patient wearing eye protection?
- Slit-lamp examination with fluorescein staining.
- If you suspect an epithelial defect but do not see one with fluorescein staining, repeat the fluorescein application to the eye. Sometimes the defect is slow to take up the dye. If the entire epithelium sloughs off, only Bowman membrane remains, which may take up fluorescein poorly.
- Eyelid eversion to search for foreign bodies.
- Evaluate for conjunctival ulcerations/defects.
- Check the intraocular pressure (IOP). In the presence of a distorted cornea, IOP may be most accurately measured with a tonometer.

Grading of severity

- **Grade 1** is characterized by clear cornea (epithelial damage only) and no limbal ischemia (excellent prognosis).
- **Grade 2** shows hazy cornea but with visible iris details and less than one-third of the limbus being ischemic (good prognosis).
- **Grade 3** manifests total loss of corneal epithelium, stromal haze obscuring iris details and between one-third and half limbal ischemia (guarded prognosis).
- **Grade 4** shows opaque cornea and more than half limbal ischemia (very poor prognosis).

Non pharmacological treatment

A. Copious irrigation

- Place an eyelid speculum and instill topical anesthetic.
- Copious irrigation using saline or ringer lactate solution for at least 30 minutes. Tap water can be used in the absence of these solutions and may be more efficacious in inhibiting elevated intracameral pH than normal saline for alkali burns.
- Check pH and if not neutral, repeat irrigation.
- NEVER use acidic solutions to neutralize alkalis or vice versa.

B. Double-eversion of the upper eyelid:

- It should be performed so that any retained particulate matter trapped in the fornices is identified and removed with cotton tip applicator.
- Calcium hydroxide particles may be more easily removed with a cotton-tipped applicator soaked in disodium ethylenediaminetetraacetic acid (EDTA).

C. Debridement: Necrosed area of conjunctiva and cornea should be debrided to promote healing.

D. Hospitalization: It is required for severe injuries to ensure regular eye drop instillation.

Pharmacological treatment

The main aims of treatment of chemical burns are to reduce inflammation, promote epithelial regeneration and prevent corneal ulceration.

- Steroids
  - It reduces inflammation. However, they also impair stromal healing by reducing collagen synthesis and inhibiting fibroblast migration.
  - So topical steroids may be used initially (usually 4–8 times daily, strength depending on injury severity) but must be tapered off after 7–10 days when sterile corneal ulceration is most likely to occur.
  - Steroids may be replaced by topical NSAIDs, which do not affect keratocyte function.
- Cycloplegia may improve comfort.
- Topical antibiotic drops are used for prophylaxis of bacterial infection
- Ascorbic acid

  Improves wound healing and promotes the synthesis of mature collagen by corneal fibroblasts. Topical sodium ascorbate 10% is given 2-hourly in addition to a systemic dose of 1–2 g vitamin C (L-ascorbic acid) QID (not in patients with renal disease).
- Tetracyclines should be considered if there is significant corneal melting and can be administered both topically (tetracycline ointment QID) and systemically (doxycycline 100 mg BD tapering to once daily).
- Symblepharon should be prevented by lysis of developing adhesions with a sterile glass rod or damp cotton bud.

Surgical treatment

- Early surgery may be necessary to promote revascularization of the limbus, restore the limbal cell population and re-establish the fornices.
- Keratoplasty should be delayed for at least 6 months and preferably longer to allow maximal resolution of inflammation.
- Keratoprosthesis may be required in very severely damaged eyes because the results of conventional grafting are poor.

CORNEAL AND CONJUNCTIVAL FOREIGN BODIES IN EYE

Salient features

- Symptoms include foreign body sensation, watering and redness with history of trauma
- Conjunctival or corneal foreign body with or without rust ring, conjunctival injection, eyelid edema, mild anterior chamber reaction, and superficial punctuate keraptopathy.

Investigations

- Determine the mechanism of injury [e.g., metal striking metal, power tools or weed-whackers may suggest an intraocular foreign body (IOFB)]. Attempt to determine the size, weight, velocity, force, and shape of the object. Document visual acuity before any procedure is performed.

- Examine closely for iris tears and transillumination defects, lens opacities, AC shallowing, and asymmetrically low IOP.
- If there is no evidence of perforation, evert the eyelids and inspect the fornices for additional foreign bodies.
- Dilate the eye and examine the posterior segment for a possible intraocular foreign body
- Avoid magnetic resonance imaging (MRI) if there is a history of possible metallic foreign body.

Treatment

Corneal foreign body

- Apply topical anesthetic (e.g., proparacaine 0.5% eye drops).
- Remove the corneal foreign body with a cotton tip applicator, 26 no needle or fine forceps at a slit lamp.
- Multiple superficial foreign bodies may be more easily removed by irrigation.
- Remove the rust ring as completely as possible on the first attempt. This may require an ophthalmic drill. It is sometimes safer to leave a deep, central rust ring to allow time for the rust to migrate to the corneal surface, at which point it can be removed more easily.
- Measure the size of the resultant corneal epithelial defect.
- Treat epithelial defect with antibiotics and carboxy methyl cellulose (CMC) eyedrops, if necessary pad it.
- Alert the patient to return as soon as possible if there is any worsening of symptoms.

Conjunctival foreign body

- Remove foreign body under topical anesthesia.
  - Multiple or loose foreign bodies can often be removed with saline irrigation.
  - A foreign body can be removed with a cotton-tipped applicator soaked in topical anesthetic or with fine forceps.
  - For deeply embedded foreign bodies, consider pretreatment with a cotton-tipped applicator soaked in phenylephrine 2.5% to reduce conjunctival bleeding.
  - Small, relatively inaccessible, buried subconjunctival foreign bodies may sometimes be left in the eye without harm unless they are infectious or proinflammatory. Occasionally, they will surface with time, at which point they may be removed more easily.
- Sweep the conjunctival fornices with a glass rod or cotton-tipped applicator soaked with a topical anesthetic to catch any remaining pieces.
- A topical antibiotic (e.g., bacitracin ointment BD; trimethoprim/polymyxin B or moxifloxacin/ciprofloxacin/ofloxacin drops QID) may be used.
- Artificial tears may be given for irritation.

Follow-up

- Corneal foreign body: Follow up as with corneal abrasion. If residual rust ring remains, reevaluate in 24 hours.
- Conjunctival foreign body: Follow up as needed, or in 1 week if residual foreign bodies were left in the conjunctiva.

TRAUMATIC HYPHEMA

Salient features

- The patient presents with blurred vision, history of blunt trauma and pain in eye.
- Blood or clot or both can be observed in the anterior chamber, usually visible without a slit lamp.
- A total (100%) hyphema may be black or red. When black, it is called an "8-ball" or "black ball" hyphema.
- When red, the circulating blood cells may settle with time to become less than a 100% hyphema.

Investigations

- History of traumatic event including force, velocity, type, and direction of injury should be taken.
- Use of medications with anticoagulant properties [aspirin, NSAIDs, warfarin or clopidogrel], personal or family history of sickle cell disease/trait, symptoms of coagulopathy (e.g., bloody nose-blowing, bleeding gums with tooth brushing, easy bruising)
- Ocular examination, first ruling out a ruptured globe. Document the extent (e.g., measure the hyphema height) and location of any clot and blood.
- Measure the IOP. Perform a dilated retinal evaluation without scleral depression.
- Consider a CT scan of the orbits and brain (axial and coronal views, with 1- to 3-mm sections through the orbits) when indicated (e.g., suspected orbital fracture or IOFB, loss of consciousness).

Treatment

- Confine either to bed rest with bathroom privileges or to limited activity. Elevate head of bed to allow blood to settle.
- Place a shield (metal or clear plastic) over the involved eye at all times. Do not patch because this prevents recognition of sudden visual loss in the event of a rebleed.
- Atropine 1% solution BD to TDS or scopolamine 0.25% BD to TDS.
- No aspirin-containing products or NSAIDs.
- Mild analgesics only (e.g., acetaminophen). Avoid sedatives.
- Use topical steroids (e.g., prednisolone acetate 1% four to eight times per day) if any suggestion of iritis (e.g., photophobia, deep ache, ciliary flush), evidence of lens capsule rupture, any protein (e.g., fibrin), or definitive white blood cells in anterior chamber. Reduce the frequency of steroids as soon as signs and symptoms resolve to reduce the likelihood of steroid-induced glaucoma.
- For increased IOP:
  - Start with a beta-blocker (e.g., timolol 0.5% or levobunolol 0.5% BD).

  - If IOP still high, add topical alpha-agonist (e.g., apraclonidine 0.5%, or brimonidine 0.2% TDS) or topical carbonic anhydrase inhibitor (e.g., dorzolamide 2%, or brinzolamide 1% TDS). Avoid prostaglandin analogs and miotics (may increase inflammation). In children under 5, topical alpha-agonists are contraindicated.
  - If topical therapy fails, add acetazolamide (500 mg orally, BD for adults, 20 mg/kg/day divided three times per day for children) or mannitol [1 to 2 g/kg intravenously IV over 45 minutes once a day]. If mannitol is necessary to control the IOP, surgical evacuation may be imminent.
- If hospitalized, use antiemetics for severe nausea or vomiting [e.g., prochlorperazine 10 mg IM TDS or 25 mg BD].
- Indications for surgical evacuation of hyphema:
  - Corneal stromal blood staining.
  - Significant visual deterioration.
  - Hyphema that does not decrease to <50% by 8 days [to prevent peripheral anterior synechiae (PAS)].
  - IOP >60 mm Hg for >48 hours, despite maximal medical therapy (to prevent optic atrophy).
  - IOP >25 mm Hg with total hyphema for >5 days (to prevent corneal stromal blood staining).

Follow up

- The patient should be seen daily for 3 days after initial trauma to check visual acuity, IOP, and for a slit-lamp examination.
- Look for new bleeding, increased IOP, corneal blood staining, and other intraocular injuries as the blood clears (e.g., iridodialysis; subluxated, disclocated, or cataractous lens). Hemolysis, which may appear as bright red fluid, should be distinguished from a rebleed, which forms a new, bright red clot. If the IOP is increased, treat as described earlier.
- The patient should be instructed to return immediately if a sudden increase in pain or decrease in vision is noted (which may be symptoms of a rebleed or secondary glaucoma). If a significant rebleed or an intractable IOP increase occurs, the patient should be hospitalized.
- After the initial close follow-up period, the patient may be maintained on a long-acting cycloplegic (e.g., atropine 1% solution OD to BD, scopolamine 0.25% OD to BD), depending on the severity of the condition. Topical steroids may be tapered as the blood, fibrin, and white blood cells resolve.
- Glasses or eye shield during the day and eye shield at night. As with any patient, protective eyewear (polycarbonate or Trivex lenses) should be worn any time a significant potential for an eye injury exists.
- The patient must refrain from strenuous physical activities (including bearing down or valsalva maneuvers) for 1 week after the initial injury or rebleed. Normal activities may

be resumed 1 week from the date of injury or rebleed. This period should be extended if blood remains in the anterior chamber.

- If patient was hospitalized, see 2 to 3 days after discharge. If not hospitalized, see several days to 1 week after initial daily follow-up period, depending on severity of condition (amount of blood, potential for IOP increase, other ocular or orbital pathologic processes).

DOG BITE (RABIES)

Rabies can be transmitted by dog bite or following non-bite exposure to mucous membrane or an open wound (one that has bled within 24 hours) with saliva. Bites from dogs in developing countries should always be considered high risk, and prophylaxis should be given without waiting for test results.

Salient features

- Prodromal symptoms - such as headache, malaise, sore throat and fever last about 3-4 days. Pain and tingling at the bitten site.
- Stage of excitation - patient is intolerant to noise, bright light or a cold draught. Aerophobia may be present. Hydrophobia is a characteristic symptom of rabies. Examination shows increased reflexes, dilatation of pupils, increased sweating, lacrimation and salivation. Mental changes include fear of death, anger, irritability and depression.
- Convulsions may occur resulting in death, paralysis and coma. The total duration of illness lasts for 2-3 days.

Guidelines for Determining Need for Rabies Prophylaxis (Adapted from Human rabies prevention—United States, 1999. Recommendations of the Advisory Committee on Immunization Practices)

Animal type & evaluation	**Post-exposure prophylaxis recommendations**
• Dogs, cats, and ferrets (healthy and available for 10-days) period*	Do not start vaccination unless animal develops clinical signs of rabies during observation
• Rabid or suspected rabid	Immediately vaccinate.
• Unknown (escaped)	Consult public health officials.

* 10-days observation period, begin postexposure prophylaxis at the first sign of rabies. If the animal exhibits clinical signs of rabies, it should be euthanized immediately and tested.

Treatment of dog bite and Post-exposure prophylaxis of rabies

Three steps: wound care, passive immunization, and active immunization.

Immediate measures

- Early treatment is essential (less than 3 hours after the bite/injury occurred).

- Scrub the wound and its edges with soap and water.
- If the wound is a puncture, swab deeply in the wound and around its edges.
- Follow with application of a virucidal agent: 1% or 2% benzalkonium chloride *or* povidone-iodine and rinse.
- Tetanus prophylaxis should be given as indicated by patient's current immunization status.
- Do not suture bite wounds immediately. If suturing is required, hold it for 24-48 hours, applying minimum number of stitches under the cover of antirabies immunoglobulin locally.

Non immunized persons

- Human rabies immuno globulin (HRIG)- full dose should be infiltrated around wound(s), and the remaining volume should be administered IM at an anatomic site distant from vaccine administration. Do not administer vaccine in the same syringe as HRIG may partially suppress active production of antibody.
- Vaccine Human diploid cell vaccine (HDCV) injected IM (deltoid area), one each on days 0, 3, 7, 14, and 28. The vaccine should be administered in the deltoid rather than the gluteal region, to avoid accidental injection into fat, which will prevent antibody formation.
- Rabies vaccine adsorbed (RVA) and purified chick embryo culture (PCEC) vaccine-administered for prophylaxis by the intramuscular route and should have antibody titers checked at 2 to 4 weeks. It is associated with fewer hypersensitivity reactions in patients receiving boosters and prophylaxis during pregnancy does not increase in fetal wastage, congenital defects, should not be withheld when indicated.
- Corticosteroids, antimalarials, and other immunosuppressives can interfere with the development of active immunity and should be withheld during the course of treatment if possible.
- Patients with a history of hypersensitivity should nevertheless be cautiously given immunoprophylaxis in a controlled setting, with antihistamines and epinephrine available

Previously immunized persons

- HRIG should not be administered.
- Vaccine HDCV or PCECV 1.0 ml, IM (deltoid area), on days 0 and 3.

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3. http://www.cdc.gov/rabies/exposure/types.html; 2008.
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SNAKE BITE

India is estimated to have the highest snakebite mortality in the world, with WHO estimates placing the number of bites at 83,000 per annum with 11000 deaths. Most of the fatalities are due to the victim not reaching the hospital in time where definite treatment can be administered.

Snakes in India

There are about 236 species of snakes in India, most of which are non-poisonous, however, there are 13 known species that are poisonous and four of these, namely Indian cobra, Russell's viper, saw- scaled viper and Indian krait are highly venomous and are believed to be responsible for most of the poisonous bites in India

Salient features

- Viper envenomization (haemovasculotoxic): swelling and local pain, tender enlargement of local lymph nodes, bleeding from any orifice,epistaxis, vomiting, hypotension, petechiae, purpura, ecchymoses
- Elapid envenomization (neurotoxic): swelling and local pain, local necrosis and blistering, descending paralysis, ptosis, diplopia, ophthalmoplegia, paralysis of jaw and tongue, numbness around the lip and mouth, inadequate ventilation, paradoxical respiration.

Investigations

- Identification of snake, local swelling within few minutes after bite is an important sig of viper envenomization.
- Blood stained sputum and non clotting of blood are diagnostic criteria for Russel viper and Saw scaled viper poisoning.
- Ptosis and glossopharyngeal palsy are diagnostic criteria for Cobra and Indian Krait poisoning.
- Investigations include
  - 20 minute whole blood clotting test
  - Hb/ platelet count/ peripheral smear/ PT/ APTT/ FDP/ D-Dimer
  - Urine examination for proteinuria/RBC/ hemoglobinuria/ myoglobinuria
  - Serum creatinine/ urea/ potassium
  - Oxygen saturation/ PR/BP/ RR/ postural blood pressure
  - ECG/ X-Ray/ CT/ ultrasound (the use of X-Ray and ultrasound are of unproven benefit, apart from identification of bleeding in viperine bites).
  - ELISA to confirm snake species

Treatment

First aid measures (RIGHT approach)

R: reassure patient and control anxiety

I: immobilize bitten part by splinting the limb

G-H: get to hospital- transfer the victim to the nearest treatment facility

T: tell the doctor about any symptoms

Do Not*: Apply tourniquet, wash the bite site with soap and any other solution, make cuts or incision on or near bitten area, apply extreme cold or herbal and folk remedies, suck venom, give the victim drink, alcohol or drugs*

- Ensure airway, breathing and shock management
- Pain can be relieved with non sedating analgesic- oral paracetamol
- Anti-snake venom (ASV) is the mainstay of treatment. In India polyvalent ASV is available which is effective against all the four common species. ASV should be administered only when there are definite signs of envenomation i.e. coagulopathy or neurotoxicity
- Mild and severe envenomation requires 8 to 10 vials. Each vial is 10 ml of reconstituted ASV. Children should receive the same ASV dosage as adults. ASV should be administered over 1 hour at constant speed and patient should be closely monitored for 2 hours.
- ASV reactions: Anaphylaxis may be life-threatening. if anaphylaxis is evident, ASV should be discontinued. Inj. hydrocortisone 100 mg + antihistamine or Inj. adrenaline subcutaneously can be administered for ASV reaction
- Repeat dose of 5-10 vials of ASV i.e. half to one full dose, should continue 6 hourly till coagulation is restored unless a species is identified against which polyvalent ASV is ineffective. Only for snakes which inject massive amounts of venom, such as the King Cobra or Australian elapids, massive doses of 50+ vials are required.

Follow up

A snakebite victim discharged within 24 hrs should be advised to return if there is any worsening of symptoms or signs such as bleeding, pain or swelling at the site of bite, difficulty in breathing, altered sensorium etc. Explain about the signs and symptoms of serum sickness which may manifest after 5-10 days.

INSECT AND ARACHNID BITES

Insect arachnid (spider, scorpion) bites and stings are common and usually cause minor irritation. However, some stings can be painful and trigger a serious allergic reaction.

Salient features

- Skin around the bite becomes red, swollen and itchy. The venom from a sting often causes a swollen, itchy, red mark (a weal) to form on the skin. This can be painful and itchy for a few days, but is harmless in most cases. They can usually be treated at home.
- The severity of bites and stings varies depending on the type of insect and sensitivity of the person. In rare cases, some people can have a serious allergic reaction (anaphylaxis) to a bite or sting that requires immediate medical treatment.

Non pharmacological treatment

- Remove a sting- by scraping it out with fingernails. Be careful during removing the sting, prevent further spread the venom into the skin and do not puncture venomous sac.
- Minor bites and stings can be treated by washing the affected area with soap and water, placing a cold compress (a cloth cooled with cold water) over the affected area to reduce swelling
- Do not scratch the area because it can become infected.
- Wrap an ice pack in a towel and place it on the swelling

Pharmacological treatment

If the bite or sting is painful or swollen

- Tab. paracetamol or ibuprofen(children under 16 years old should not be given aspirin)
- Apply a spray or cream that contains local anesthetic, antihistamine or mild hydrocortisone (1%) on the affected area to prevent itching and swelling
- Tab. cetrizine 10 mg once a day and topical administration of calamine lotion to relieve pruritus
- Systemic corticosteroids may be appropriate if there are severe side-effects.
- Any person who collapses, or complains of wheezing, feeling of anxiety or faintness, generalized itching, or tightness in the chest within 1 hour of being stung by an insect should be treated as having anaphylactic shock. Inj. adrenaline 1 mg (as hydrogen tartrate) 0.5-1.0 ml IM injection (1:1000 solution), repeat every 15-20 min if required. All patients should be observed at least for 24 hours for recurrent anaphylaxis (see section on anaphylactic shock).

SCORPIONS BITE

- Simple analgesics, such as paracetamol and aspirin, can be given to relieve pain. However, because of the potential for severe reactions, every effort should be made to get the patient to a hospital as soon as possible.
- Vasodilators, administered in a hospital setting within 24 hours of the attack, may attenuate the cardiovascular response and possibly reduce mortality.
- In endemic area, species-specific antiscorpions sera may be available locally and this can be of value if administered within few hours.

Reference

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Chapter 4

CARDIOVASCULAR DISEASES

INFECTIVE ENDOCARDITIS

It is a microbial infection of endothelial surface of heart. Acute bacterial endocarditis is caused by *S. aureus*, *group A hemolytic streptococci, pnuemococci or gonococci*. Sub acute bacterial endocarditis is caused by virulent *S. Viridans* or other *streptococci.*

Salient features

- Symptoms include fever, night sweats, weight loss, anorexia, cough, dyspnoea, arthralgia, chest pain.
- Signs are murmur, splenomegaly, clubbing, embolic feature and signs of immune complex deposition, like Janeway lesions, Roth spots, splinter hemorrhages

Pharmacological treatment

- Empirical therapy: Pending the availability of diagnostic data, blood culture–negative sub acute native valve endocarditis is treated either with Inj. ampicillin-sulbactam 12 g every 24 h or with Inj. ceftriaxone 1 gm OD plus Inj. gentamicin 3 mg/kg QID IV; Tab. doxycycline 100 mg twice daily is added for *Bartonella* coverage.
- After blood culture report organism specific antibiotic therapy are as under *Streptococci* :
  - Penicillin sensitive: Inj. penicillin G 2–3 MU IV q4h or Inj. ceftriaxone 2 g IV QID for 2 weeks plus Inj. gentamicin 3 mg/kg QID IV or IM, as a single dose or divided into equal doses TDS for 2 weeks.
  - Penicillin resistant: Inj. ceftriaxone 2 g IV QID for 4 weeks plus Inj. gentamicin 3 mg/kg QID IV or IM, as a single dose or divided into equal doses TDS for 2 weeks
- *Enterococci*: Inj. penicillin G 4–5 MU IV every 4 hours plus Inj. gentamicin 1 mg/kg IV TDS, both for 4–6 weeks.
- *Staphylococci*: Inj. nafcillin or oxacillin 2 g IV q4h for 4–6 weeks or Inj. cefazolin 2 g IV TDS for 4–6 weeks. If methicillin resistant: Inj. vancomycin 15 mg/kg IV 12 hours for 4 to 6 weeks.
- *HACEK group*: Inj. ceftriaxone 2 g OD IV for 4 weeks
- Prophylaxis for infective endocarditis in patients with vulnerable heart valve. Dental procedures wherein there is manipulation of gingival tissue or peri-apical regions of the teeth or perforation of the oral mucosa (including surgery on the respiratory track).

Antibiotic regimens for prophylaxis

- Tab. amoxicillin 2 g 1 hour before procedure
- Inability to take oral medication: Inj. ampicillin 2gm IV/IM 1 hour before procedure
- Penicillin allergy: Tab. clarithromycin or azithromycin 500 mg 1 hour before procedure or Tab. cephalexin 2 gm, 1 hour before procedure or Tab. clindamycin 600 mg 1 hour before procedure

- Penicillin allergy, inability to tolerate oral medication: Inj. cefazolin or ceftriaxone 1 gm IV/IM,30 min before procedure or clindamycin 600 mg IV/IM, 1 hour before procedure.

ACUTE PERICARDITIS

Inflammation of pericardium is called as pericarditis.

Salient features

- May be caused by viruses, bacteria, mycobacteria, connective tissue disorders, uremia, malignancy, radiation or trauma.
- Pleuritic chest pain which relieves on bending forward breathlessness, tachycardia
- Fever, pleural rub, elevated ESR, irregular pulse, symmetric enlargement of cardiac silhouette

Pharmacological treatment

- Tab. aspirin 650 mg orally every 3-4 hours or other NSAIDs like Tab. indomethacin 100-150 mg orally daily in divided doses are usually effective.
- Tuberculosis pericarditis: Anti tuberculosis treatment
- Uremic pericarditis: dialysis
- Usually with antibiotics patients have to be carefully observed as they may develop pericardial effusion. If cardiac tamponade develops, pericardiocentesis must be carried out at once.

CARDIOMYOPATHY

The cardiomyopathies are a group of diseases that affect the heart mused itself and are not the result of hypertension, congenital or acquired valvular, coronary or pericardial abnormalities.

Salient features

- Fever, chest pain, basal rales, pedal edema, cardiomegaly, syncope, ventricular ectopy,
- LVH, raised Jugular Venous Pressure (JVP), diffuse dyskinetic apex beat, S3, hepatomegaly, peripheral edema, murmurs of mitral and tricuspid regurgitation

Non pharmacological treatment

- Salt restriction, alcohol discontinuation

Pharmacological treatment

Standard therapy for heart failure with following drugs:

- Diuretics- hydrochlorothiazide 6.25–50 mg or chlorthalidone 25–50 mg, loop diuretics- furosemide 40–80 mg, aldosterone antagonists- spironolactone 25–100 mg
- Beta blockers: atenolol 25–100 mg, metoprolol 25–100 mg, propranolol 40–160 mg
- ACE inhibitors: captopril 25–200 mg**,** lisinopril 10–40 mg**,** ramipril 2.5–20 mg

- Angiotensin II antagonists: losartan 25–100 mg, valsartan 80–320 mg, candesartan 2–32 mg
- Calcium antagonists: nifedipine (long-acting) 30–60 mg, verapamil (long-acting) 120–360 mg, diltiazem (long-acting) 180-420 mg

Hypertrophic cardiomyopathy

- β blockers is the first line of drug.
- Calcium channel blockers, especially verapamil, have also been effective in symptomatic patients. Diuretics for heart failure
- Dual chamber pacing for refractory cases
- Nonsurgical septal ablation with alcohol injection in septal branches.

Drugs to be avoided

- Calcium channel blockers and NSAIDs – in dilated cardiomyopathy
- Digitalis, Nitrates, Vasodilators, B-agonists- avoided in Hypertrophic obstructive cardiomyopathy (HOCM)

HYPERTENSION

Usually asymptomatic and discovered on routine measurement of blood pressure. Secondary hypertension (HT) presents as a part of a symptom complex as in acromegaly, Cushing's disease, renovascular or renal parenchymal disease, connective tissue disorders (SLE, scleroderma, etc.), or coarctation of aorta.

Salient features

- Nonspecific symptoms are fatigue, headache, epistaxis.
- Uncontrolled hypertension can lead to target organ damage (TOD) such as coronary artery disease (CAD), left ventricular hypertrophy (LVH), cerebrovascular accidents (CVA), transient ischemic attacks (TIA), retinopathy, peripheral vascular diseases including dissecting aneurysm, renal disease.
- Associated risk factors are - age >55 years in males and >65 years in females, smoking, diabetes mellitus, microalburninuria or GFR <60 ml/hr, hyperlipidemia, family history, obesity, sedentary life-style and ethnic group.

Investigations

Routine laboratory tests recommended before initiating therapy include,

- Electrocardiogram
- Urinalysis;
- Blood glucose and hematocrit; serum potassium, creatinine (or the corresponding estimated GFR, and calcium;
- Lipid profile (after a 9-12 hour fast) that includes high-density lipoprotein, cholesterol, low-density lipoprotein cholesterol, and triglycerides.

- Optional tests include measurement of urinary albumin excretion or albumin/creatinine ratio. More extensive testing for identifiable causes is not indicated generally unless BP control is not achieved.

Precautions to be taken while measuring BP

- The auscultatory method of BP measurement with a properly calibrated and validated instrument should be used. BP should be measured using an appropriate size cuff in both upper limbs and at least one lower limb in both supine and erect posture.
- Patient should have been resting for at least 5 minutes and should not have consumed coffee, smoked during the last 30 minutes before measuring the blood pressure. At least 2 measurements should be made. Systolic BP is the point at which the first of 2 or more sounds are heard (phase 1) and diastolic BP is the point before the disappearance of sounds (phase 5).

Non pharmacological treatment

- Reduce dietary sodium intake to no more than 100 mmol per day (2.4 g sodium or 6 g sodium chloride).
- Adopt DASH eating plan- diet rich in fruits, vegetables, with low fat dairy products with a reduced content of saturated and total fat.
- Life-style modification - exercise consisting of vigorous aerobic exercise like brisk walking, swimming or jogging at least 20 to 30 minutes on most days of the week with the heart rate reaching 65-70% of maximal heart rate.
- Weight control using a combination of dietary and exercise measures to maintain normal body weight (body mass index 18.5-24.9 kg nr).
- Moderation of alcohol consumption - limit alcohol to no more than 2 drinks (1 oz or 30 ml ethanol; e.g., 24 oz beer, 10 oz wine, or 3 oz 80-proof whiskey) per day in most men and to no more than 1 drink per day in women and lighter weight persons**.**
- Cessation of smoking.
- Yoga

Pharmacological treatment

- Pre – hypertension: behavioral modification follow up
  Stage 1: thiazide diuretics OR calcium channel blockers or beta blockers or ACEI
  Stage 2: Combination therapy (dosages to be adjusted according to blood pressure response)
- Diuretics preferred in obese, elderly, CCF
- Beta blockers in young, coronary artery disease (CAD), vascular headache, atrial fibrillation
- Calcium channel blockers in elderly, CAD, ACEI in diabetic, LVF, young
- Alpha blockers: prostatitism, dyslipidemia
- Alpha methyl dopa- pregnancy

Dose of commonly uses antihypertensive drugs

- Tab. hydrochlorothiazide 12.5 -25 mg OD or BD
- Tab. frusemide 20-40 mg twice day
- Tab. frusemide 20 + spironolactone 50 mg 1-2 times day
- Tab. indapamide 2-5 mg daily
- Tab. atenolol 25-100 mg daily or Tab. metoprolol 25-150 mg BD
- Tab. amlodipine 2.5- 20 mg OD or Cap. nifedipine 20-80 mg as sustained release daily
- Tab. enalapril 2.5 mg OD or Tab. lisinopril 2.5 mg OD or Tab. ramipril1.25-10 mg OD
- Tab. losartan 25-50 mg OD or BD
- Tab. prazosin 1-20 g OD at bed time
- Tab. clonidine 0.05-0.6 mg BD or Tab. methyldopa 250-1000 mg OD

Accelerated hypertension

Patients presenting with BP 200/140 mm Hg or more without papilledema

- Inj. enalapril 1.25-5 mg IV 6 hourly Or Inj. nitroprusside 0.25-1.0 mcg/Kg/min IV infusion (dose to be titrated with BP, maximum dose for 10 meg only) Or
- Inj. nitroglycerine 5-100mcg/min infusion Or
- Inj. hydralazine 10-20 mg IV or IM, if no response at 20 mg change the drug Or
- Inj. labetalol 20-80 mg IV every 5-10 min up to a total of 300 mg, infusion 0.5-2 mg/min, oral 100-600 mg 2 times a day. Or
- Inj. phentolamine 5-15 mg IV (especially useful in pheochromocytoma)

Malignant hypertension

- Patients presenting with BP 200/140 mm Hg and evidence of vascular damage like papilledema, deranged renal function, should be referred to a tertiary care centre.

Resistant hypertension

- Resistant hypertension is the failure to reach goal BP in patients who are adhering to full doses of an appropriate 3-drug regimen that includes a diuretic. Refer these patients to hypertension specialist.

Follow up

- Regular BP monitoring to ensure optimal or normal BP in young or middle aged or diabetic subjects (130/80 mm Hg) and at least high normal BP in elderly patients (140/90 mm Hg).
- Monitor all risk factors for coronary artery disease.
- Ensure compliance with life-style modification.
- Assess for complications and target organ damage.
- In accelerated or malignant HT, avoid sudden fall in diastolic BP to below 100 mm Hg

Hypertension in children and adolescents

- In children and adolescents, hypertension is defined as BP that is, on repeated measurement, at the 95th percentile or greater adjusted for age, height, and gender. The fifth Korotkoff sound is used to define diastolic BP.
- Be alert for possibility of identifiable causes of hypertension in younger children (i.e., kidney disease, coarctation of aorta).
- Life style interventions are strongly recommended, with pharmacological therapy instituted for a higher levels of BP or if there is insufficient response to lifestyle modification.
- Choices of antihypertensive drugs are similar in children and adults, but effective doses for children are often smaller and should be adjusted carefully.

Patient education

- Explain consequences of uncontrolled HT and the need for the long-term control with medication.
- Advice regarding control of other risk factors likes diabetes, hyperlipidemia, etc.
- Diuretics should be taken in the morning and if two doses a day are required second dose should be given before 4 PM.
- Manage target organ damage - referral to a higher centre for CAD (PTCA/ CABG), nephropathy (prepare for dialysis or transplant), carotid doppler to determine presence of atheroma and possibility of end arterectomy

ANGINA PECTORIS

It is clinically syndrome due to myocardial ischemia, caused by critical obstruction in coronary arteries due to atherosclerosis, calcific arotic stenosis, or due to spasm or emboli.

Salient features

- Precordial chest pain due to stress or exertions, palpitation, breathlessness symptoms are typically associated with exertion and emotional upset and are relieved quickly by rest and/or sublingual nitrates ECG shows ST depression, T wave inversion, LVH.

Non- pharmacological treatment

- Avoid heavy exertion, take rest during acute stage, stop smoking, weight reduction.

Pharmacological treatment

- Tab. isosorbide dinitrate 5 mg sublingual during the attack, dose repeated as required or Tab nitroglycerine 0.3-0.6 mg sublingual
- If attack occurs twice in week, chronic prophylaxis is recommended with Tab. metoprolol 50 mg BD or Tab. atenolol 50 mg OD
- If beta blockers are contraindicated or angina persists- Tab. diltiazem- 60-120 mg/day
- Tab. aspirin 100-150 mg/ day or Tab. clopidogrel 75 mg/day

MYOCARIDAL INFRACTION

Myocardial Infarction (MI) is due to the formation of occlusive thrombus at the site of rupture of an atheromatous plaque in a coronary artery.

Salient features

- Chest pain, breathlessness, perspiration, syncope, tachy or bardycardia, hypo or hypertension, sudden cardiac arrest, ECG changes showing ischemia or infarction, raised cardiac markers

Pharmacological treatment

- Hospitalization and compete immobilization
- Oxygen – 4-6 liters / min
- Oral uncoated chewable Tab. aspirin – 325 mg
- Tab. isosorbide dinitrate – 5 mg up to 3 times sublingually
- Tab. atorvastatin – 80 mg
- Tab. clopidogrel – 300 mg
- Inj. morphine sulphate 10 mg or Inj. diamorphine 5 mg IV
- Consider thrombolysis with intravenous streptokinase or alteplase. Thrombolysis to be done within 30 mins of time of admission.
- Inj. streptokinase 1.5 million units in 100 ml of saline as an intravenous infusion over 1 hour**.** The drug may cause hypotension, which can be managed with fluids and restricting the infusion at a slower rate. Or
- Inj. alteplase - rTPA (Recombinant tissue plasminogen activator): The regimen is given over 90 minutes (bolus dose of 15 mg followed by 0.75 mg/kg of body weight and then 0.5 mg/kg body weight over 60 minutes)
- Inj. metoprolol 5 mg IV every 2-5 minutes for 3 doses and changed to Tab. metoprolol 50 mg every 6 hours for 48 hours, followed by 100 mg every 12 hours.
- Inj. Nitroglycerine IV 0.6 – 1.2 mg/hour
- Inj. heparin 200 units/Kg SC twice daily for 5 days is given in addition to oral aspirin may prevent reinfarction after successful thrombolysis and reduce the risk of thromboembolic complications.
- Monitor ECG, blood pressure and cardiac enzymes

Contraindications to thrombolytic therapy

- Absolute contraindications are active internal bleeding, previous sub-arachnoid or intra-cerebral hemorrhage, non hemorrhagic stroke within the last 1 year, uncontrolled hypertension (systolic > 180mm Hg, diastolic > 110mmHg)
- Relative contra-indications are current use of anti-coagulants, recent (< 2 weeks) invasive surgical procedure, recent prolonged (> 10 min) cardiopulmonary resuscitation, known bleeding diathesis, pregnancy, hemorrhage prone ophthalmic conditions, active peptic ulcer

disease, recent history of severe hypertension. Notes: Streptokinase should not be used, if given in the preceding 5 days to 2 years because of the risk of allergic reactions.

CONGESTIVE HEART FAILURE (CHF)

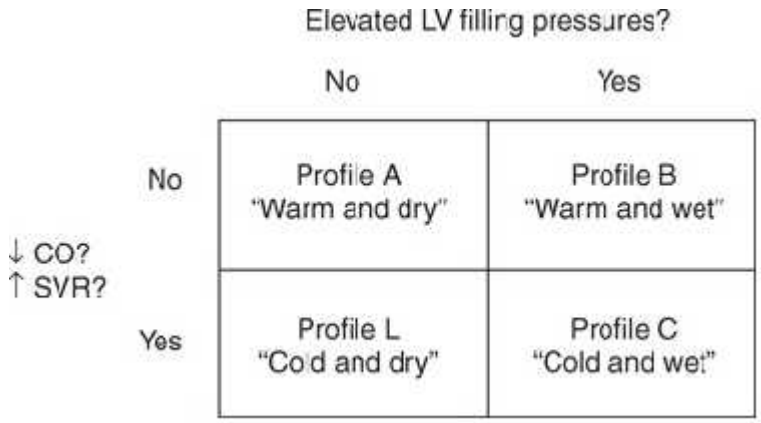
CHF is critical syndrome of inadequate cardiac output resulting in fluid retention in the lungs, abdominal organs and peripheral tissue.

Salient features

- Cough expectoration, breathlessness, pedal edema, hypo or hyper tension, easy fatigability.
- Raised jugular venous pressure, positive abdominojugular reflex, basal rales, pulmonary edema, orthopnea, paroxysmal nocturnal dyspnea (PND). S3, hepatomegaly, ascites, sinus tachycardia**.** Chest X ray shows cardiomegaly, pulmonary congestion.

Patients of heart failure are divided in 4 categories for management (Figure 1).
For category A – No further treatment needed
For category B – Diuretics and vasodilators
For category C – Ionotropic agents
For category L– Fluid challenge can be given

Figure 1. Categories of heart failure for ease of management



Non pharmacological treatment

- Restrict physical activity, bed rest in propped up position with a back rest. Dietary salt (sodium) restriction (2 – 3 g/day).
- Avoid and discontinue NSAIDs, high dose Beta blockers and calcium channel blocker. Fluid restriction depending on output and other conditions.

Pharmacological treatment

- Treatment consists of judicious mix of vasodilators, diuretics and inotropic support
- Inj. furosemide 40 – 80 mg IV stat every 2 to 3 hours with maximum up to 200 mg/day. Maintenance dose is 40 mg IV 12 hourly.

- Tab. spironolactone 25 – 200 mg/day or Tab. chlorthiazide 250 – 500 mg/day or Tab. indapamide 2.5 – 5 mg/day or Tab. benzthiazide 25 mg plus Tab. triamterene 50 mg/day.
- Tab. enalapril 2.5 -20 mg/day or Tab. lisinopril 2.5 – 10 mg/day
- Tab. isosorbide mononitrate 60 mg/day as slow release preparation at night
- Digoxin is indicated in fast ventricular rate (e.g. atrial fibrillation) and LVEF < 30 %. Tab. digoxin 0.5 mg first day followed by 0.25 mg/day 5 days/week
- Tab. carvedilol 3.15 – 25 mg per day in one or two divided doses is useful in all symptomatic or asymptomatic heart failure and in case of reduced ejection fraction (40%)
- Inj. heparin 5000 IU 12 hourly SC if patients is bedridden

Monitoring of CHF patient

- Strict intake – output charting
- Daily weight and abdominal girth measurement
- Serum electrolytes and uric acid

ARRHYTHMIAS

Arrhythmias are divided into supraventricular tachycardia, ventricular tachycardia, bradyarrhythmias

Salient features

- Palpitation**,** breathlessness**,** syncope**,** chest pain**,** "gabharaman"**,** visual disturbances**.** tachy or bradycardia**,** hypo or hypertension**.** ECG changes according to type of arrhythmia

SUPRAVENTRICULAR TACHYCARDIA

Pharmacological treatment

- Depends on the hemodynamic status of the patient.
- If he is haemodynamically stable:
  - Treatment of acute episode includes vagal maneuvers (carotid sinus massage, valsalva maneuver)
  - If the measure fails, AV nodal blocking drugs are tried. Inj. adenosine 6 mg IV bolus followed by 10 ml saline flush is given. 12 mg IV bolus can be repeated if no response after 2 minutes. Contraindications include sick sinus syndrome or second or third degree AV block.
  - Inj. metoprolol 5 mg IV can be repeated after 5 minutes or
  - Inj. verapamil 5-10 mg IV bolus over 2-3 min. can be repeated after 15 mins or Inj. diltiazem 0.25 mg/kg IV bolus over 2 min. followed by IV infusion at a rate of 10mg/hour.
  - If haemodynamically unstable, prompt synchronized direct current (DC). Cardioversion (using 10 to 100 joules) has to be done.

Maintenance therapy

Once sinus rhythm is established, chronic maintenance therapy includes,

- Tab. diltiazem 30 mg TDS or Tab. verapamil - 40 mg TDS OR
- Tab. metoprolol 25-100 mg BD or Tab. atenolol 25-100 BD OR
- Tab. digoxin 0.125-0.5 mg OD

Curative therapy

For supraventricular tachycardia involving accessary pathway is radio-frequency ablation.

VENTRICULAR TACHYCARDIA

Runs of three or more consecutive idioventricular beats exceeding a rate of 100 beats per minute.

Types:

- Sustained – If ventricular tachycardia (VT) persists for > 30 seconds.
- Non-sustained – 3 beats to 30 seconds.
- Monomorphic - Configurations uniform
- Polymorphic – Configuration varies from beat to beat

Pharmacological treatment

1. Patients with nonsustained arrhythmias, without organic cardiac disease are asymptomatic- need not be treated.
2. Patients with sustained VT in the absence of cardiac disease are symptomatic and require therapy with Tab. amiodarone 200-400 mg /day TDS
3. In patients with VT with organic cardiac disease, if haemodynamic instability, unsynchronized the arrhythmia.
4. If haemodynamically stable, the following drugs can be administered:
5. Inj. lidocaine – 1 mg/kg IV bolus, 0.5-1 mg/kg boluses may be repeated at 5 min intervals followed by 2-4 mg/min maintenance Infusion.

- Inj. amiodarone may be tried.
- If these drugs fail, over drive padding has to be done.
- To prevent recurrences, sustained VT-ICD implantation is effective.

ATRIAL FIBRILLATION

Common arrhythmia, a variant of supraventricular tachycardia.

Salient features

- Palpitation, breathlessness, gabharaman, giddiness**.** Tachycardia, irregularly irregular pulse, hypotension is commoner than hypertension

Pharmacological treatment

Acute atrial fibrillation

- Treat the precipitating factors like thyrotoxicosis
- If the patient is haemodynamically unstable, electrical cardio version using 200 joules is the treatment of choice.
- If haemodynamically stable, ventricular rate control is adequate by Tab. metaprolol 25-100 mg BD
- Tab. warfarin 5 mg OD and dose adjust as per INR

Surgical treatment

- Ablation around pulmonary veins.

BRADYARRHYTHMIAS

Sinus bradycardia – regular heart rate < 60/min.

Types: sino atrial block and atrioventricular block

Pharmacological treatment

- Inj.atropine 0.6 mg IV repeated 4- 6 hourly or Inj. isoproterenol 2-10 mcg/min infusion

Surgical treatment

- Temporary pacemaker
- If needed permanent pacemaker insertion.